Development of Machine Learning Algorithms for Automatic Geotag Recognition in Textual Data

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Abstract

The need for geographic information extraction from unstructured texts has become crucial to many areas such as urban planning and disaster response. In this thesis, we compare several machine learning methods for automatic geotag recognition (which can be considered an aspect of named entity recognition [NER]) when applied to a large multilingual dataset of complaints about public transportation from citizens of Kazakhstan who speak Kazakh or Russian; this dataset was generated by City Transportation Systems (CTS) in Astana, Kazakhstan. We have implemented four different approaches to recognize the location of each record of complaints in the dataset: (1) A few-shot Large Language Model (LLM) pipeline utilizing Qwen2.5 (7B), which was deployed on-site through Ollama, (2) A few-shot Large Language Model pipeline utilized the same method as above but with a fine-tuned Multilingual-E5-base (Me5-base) encoder, (3) Fine-tuning of XLM-RoBERTa, (4) Fine-tuning of ruBERT.

There were six domain specific types in the dataset: `STREET_NAME` , `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_INTERSECTION`. Each type was tagged according to the BIO tagging scheme. Results from our experiments show that the Qwen2.5 LLM achieved the highest precision (0.788) and macro F1 score (0.635); it was also the only model able to identify the complex `STREET_INTERSECTION` tag. The Me5-base achieved the best recall (0.653). The XLM-RoBERTa had the greatest ability to identify a particular route number tag (F1 = 0.920), while ruBERT's (F1 = 0.223) lack of success in identifying code switching multilinguality data resulted in its being unable to correctly perform NER in this environment.

Keywords: Named Entity Recognition, Geotag Extraction, XLM-RoBERTa, BERT, Large Language Models, Multilingual NLP, Kazakh, Russian, Transport Complaints, BIO Tagging, Sequence Labeling

Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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List of Tables

Table **Page**

1.1 Structural characteristics of entity categories in the CTS corpus (adapted from [1]). 9
1.2 Summary of key related works on NER and geotag extraction. 18

2.1 Annotation statistics for the CTS geotag corpus. 29
2.2 Dataset statistics by entity type after stratified 70/30 split. *Presence count* is the number of records containing at least one instance of the entity type; *tokens* counts all B- and I-labelled tokens. 31
2.3 XLM-RoBERTa fine-tuning configuration. 33
2.4 Qwen2.5-7B inference configuration. 37
2.5 Summary of evaluation metrics and their operational interpretation. 41

3.1 Training configurations of the fine-tuned encoders. 45
3.2 XLM-RoBERTa entity-level test performance ($n = 825$). 46
3.3 ruBERT entity-level test performance, corrected pipeline ($n = 825$). 48
3.4 Multilingual E5-base entity-level test performance ($n = 825$). 51
3.5 Qwen2.5-7B (Ollama, few-shot) entity-level test performance ($n = 300$). . . . 52
3.6 Comparison of four open-weight LLMs under identical few-shot prompting on the 300-record evaluation subset. Macro-averaged metrics. 54
3.7 Baseline results on the test partition. 55
3.8 Final comparison of the four primary approaches on the test partition. Throughput ρ is the average number of records processed per second on a Tesla T4 GPU (encoders) or Ollama CPU runtime (Qwen2.5). For Qwen2.5, macro figures are computed on the 300-record evaluation subset; for the encoders, on the full 825-record test set. 56
3.9 Per-class F_1 across the four primary approaches. Highest score per row is in bold. 57

3.10 Boundary stability (mean character-level Jaccard, Equation (3.1), over type-matched overlapping span pairs). 62
3.11 Confusion rates on the two diagnostic axes (token-level). Rows show the percentage of gold tokens of the source class that the model labels as the destination class. 63

List of Figures

Figure **Page**

3.1 Macro-averaged F_1 on the test partition across all four primary models and the four reference baselines. The dashed horizontal line marks the operational acceptance threshold of 0.50 macro F_1 implied by the requirements of Section 1.6.2. 56

3.2 Per-class entity-level F_1 across the four primary approaches. Each group of four bars corresponds to one entity type; `ROUTE_NUM` is essentially saturated by all multilingual systems, while `STREET_INTERSECTION` is detected only by Qwen2.5. 58

Contents

Abstract **i**

Author's declaration **ii**

List of Tables **iii**

List of Figures **v**

Introduction **1**

1 Theoretical foundations and domain analysis **4**

1.1 The relevance of automatic geotag recognition in textual data 4

1.2 General characteristics of geographic named entities and toponyms in transport-domain texts 7

1.3 Modern approaches to automatic named entity recognition 10

1.3.1 Rule-based and statistical methods 10

1.3.2 Transformer-based encoders (BERT, RoBERTa, XLM-R) 12

1.3.3 Embedding-based pipelines (E5, multilingual models) 13

1.3.4 Large language models for few-shot NER 14

1.4 Using machine learning and natural language processing for multilingual text analysis 15

1.5 Comparison of NER architectures for low-resource and code-switched (Russian–Kazakh) settings 18

1.6 Setting the task 21

1.6.1 Formulation of the purpose and objectives of the thesis on automatic geotag recognition 21

1.6.2 Requirements for an intelligent geotag-extraction system 22

1.6.3 Formulation of data requirements for model training 23

1.6.4 Defining criteria for evaluating the effectiveness of NER models . 24

2 Model development and experimental setup 27
2.1 Data collection and preparation 27
2.1.1 Data collection: transport complaint records from City Transportation Systems 27
2.1.2 Manual BIO annotation in Label Studio 28
2.1.3 Preprocessing pipeline 29
2.1.4 Stratified train–test split 30
2.2 Machine learning models 31
2.2.1 Model 1: XLM-RoBERTa fine-tuned for token classification 32
2.2.2 Model 2: RuBERT (monolingual Russian BERT) 33
2.2.3 Model 3: Multilingual E5-base with token classification head . . . 34
2.2.4 Model 4: Qwen2.5-7B via few-shot prompting 35
2.3 Baseline methods 37
2.3.1 Regex-based rule extraction 37
2.3.2 Frozen XLM-R linear probe 37
2.3.3 CRF with handcrafted features 38
2.3.4 Most-frequent-class (MFC) baseline 38
2.4 Evaluation metrics 39
2.4.1 Per-class precision, recall, and F1-score 39
2.4.2 Macro- and micro-averaged F1 39
2.4.3 Boundary stability 40
2.4.4 Confusion matrix analysis 40
2.4.5 Summary of evaluation dimensions 40

3 Comparative analysis and integration of the best model 42
3.1 Experimental setup and training configurations 43
3.1.1 Unified evaluation protocol 43
3.1.2 Aggregation and auxiliary metrics 44
3.1.3 Hardware, software, and reproducibility 45
3.2 Per-model results and error analysis 46
3.2.1 XLM-RoBERTa: results and error analysis 46
3.2.2 ruBERT: results and limitations on Russian–Kazakh mixed text . 48
3.2.3 Multilingual E5-base: results and error analysis 50
3.2.4 Qwen2.5 LLM: results and analysis of few-shot prompting behaviour 52
3.3 Baseline comparison and headline results 54

3.4 Comparative analysis of the four approaches 56
3.4.1 Aggregate quality 57
3.4.2 Per-class behaviour 57
3.4.3 Robustness on rare classes 59
3.4.4 Inference cost and deployment complexity 59
3.4.5 Synthesis 60
3.5 Analysis of challenging cases 60
3.5.1 Extremely rare entities: `STREET_INTERSECTION` 61
3.5.2 Russian–Kazakh code-switching 61
3.5.3 Noisy informal input 62
3.5.4 Boundary instability on multi-word entities 62
3.5.5 Confusion-axis analysis 63
3.6 Selection of the optimal model and integration into the CTS pipeline . . 64
3.6.1 Operational requirements revisited 64
3.6.2 Decision framework 64
3.6.3 Recommended deployment: hybrid pipeline 65
3.6.4 Integration with the CTS complaint system 65
3.6.5 Limitations of the recommended deployment 66
3.6.6 Summary 66

4 Conclusion **68**

Bibliography **70**

Introduction

The increasing amount of digitalized data related to transportation networks is producing an increasing number of complaints from citizens. Because of this, public transport entities are receiving a large amount of complaint messages on a daily basis through mobile apps, web sites, and phone calls. In each message or complaint, a passenger provides their own account of an operational failure; i.e., they describe how the operation failed (e.g. a bus missed a scheduled stop, the driver deviated from the route, etc.) using free form text. Within this unformatted text, there are many geographic locations mentioned – e.g. route numbers, names of bus stops, names of streets, identifiers for vehicles and intersections, etc. – that could be used by various transportation network operators to improve real-time monitoring of service performance, optimize routes, and automate response to incidents. Thus, as both the volumes of input grow and the costs associated with manually reviewing every complaint continue to increase, there is a compelling business case for developing automated tools to identify relevant geographic information from complaint messages.

Although there are many studies that focus on named-entity recognition (NER), most NER models were trained using high-resource languages and generic domains like news articles or biomedicine. The reason why the majority of existing models do not provide sufficient detail for transport-domain operation texts from low resource bilingual settings is that they have not been precisely customized for them. The CTS complaints in Astana, Kazakhstan represent an example of this. There are approximately 3000 monthly complaint records processed by City Transportation System (CTS). However, all complaints were written in a language mix of Kazakh and Russian and had multiple instances of code-switching; informal register; numerous abbreviations relevant to transportation; and varying levels of punctuation. To date, none of the previous works targeted all these specific combinations: Kazakh-Russian code-switching; transport-domain complaint text; a six-class entity classification system, which includes several types of structurally complex composite entities; and a comparison of different machine-learning architectures applied to the same data under equivalent experimental conditions.

To meet these requirements, this study intends to develop and compare several machine learning techniques for identifying geotags automatically using multi-language tex-

tual information. As such, it is intended that the results of this study will be used as part of an operational solution for CTS.

This is achieved through the development of the study, which includes (i) examining the existing literature on named entity recognition and geotagging in relation to four paradigms – rule-based, statistical, neural and large language model – (ii) creating a dataset of 2 750 annotated public transport complaints in Kazakh and Russian languages (including pre-processing); (iii) implementing and evaluating four machine learning architectures – including three based on a pre-trained XLM-RoBERTA, two based on pre-trained DeepPavlov ruBERT, one based on Multilingual-E5-base and one based on few-shot Qwen2.5 LLM inference – and their corresponding baselines relative to precision, recall and F1 scores for each class and globally; and (iv) integrating the best performing architecture into an optimal inference pipeline for use in real time CTS complaint handling. The object of investigation is the named entity recognition within the context of public transportation complaint texts written in both Kazakh and Russian languages; the subject of the investigation is the family of machine learning models and algorithms capable of extracting six different geotag entities from noisy, code-switching public transportation complaint texts.

Chapter 1. Explanation of theoretical bases for geographic named entities in transport domain text through analysis of linguistic and operational features. This chapter summarises current approaches to automated entity recognition from rule-based, statistical methods, transformer encoders and large language models, as well as particular difficulties of Kazakh-Russian code-switching and morphological complexity. Furthermore it provides a brief summary of 21 key related works. On this basis, the chapter determines the objective of the study, defines six types of entitiy, sets out requirements for training data and formulates quantative criteria for evaluating effectiveness of model.

Chapter 2. The chapter covers model development and experimental configuration. The section outlines the methodology for collecting, annotating, and pre-processing the data; it explains how the code-switched data will be pre-processed as well as sub-word aligned for use in the four major models. It provides the details regarding the training parameters used with each of the four major models. It describes the four baselines. Finally, it discusses the metrics that were used for evaluating all of the models.

Chapter 3. This chapter is the first of two chapters that provide an overview of the results from the comparative experiments conducted on this study. The chapter includes tables which summarize the performance of each of the four models being evaluated. In addition to providing those summary tables, the chapter also contains a description

of the errors encountered during testing (e.g., rare entities; failed code-switches; noisy input) by each of the models. Finally, since multiple models are being compared within this study, there is also an overall cross-model comparison in order to provide a basis for recommending a single model for future use in the CTS production environment.

Beginning with Chapter 4, the main findings are summarized; the contribution of this study is outlined as well as potential avenues for further investigation, namely; (1) augmenting datasets with targeted data regarding rare entities, and (2), developing hybrid architecture systems that incorporate both encoders and LLMs.

This work has an applied or practical value due to its ability to directly address the needs of real world public transportation managers. The proposed system enables transport managers to rapidly and accurately perform geotagging on citizen complaint reports. This enables them to respond to incidents quickly, identify geographic "hot spots" automatically and to diminish their reliance upon manually processing these complaints by trained staff. Additionally, the annotated dataset and comparative results will serve as a starting point for future studies in which multilingual NLP meets smart city development.

Chapter 1

Theoretical foundations and domain analysis

1.1 The relevance of automatic geotag recognition in textual data

A constant flow of unstructured operational data as a result of the rapid digitalization of urban service delivery processes is being received by public transportation providers (e.g. complaints), emergency response organizations, and municipal agencies daily. These types of textual input contain valuable geographic information about where events occurred or problems were reported (street name, stop on bus/train/rail, intersection) which can be used operationally in real-time with reliable extraction. One of the most important challenges for both natural language processing and geospatial computing today is to identify and structure geographic reference in un-structured, multilingual user generated content.

Smart Cities are a growing global movement. Smart City systems aim to enhance living in cities through a combination of data-based technology and information communications technologies (ICTs) used as part of local government services. Transportation management is one of the most significant areas in which smart city initiatives have potential for large-scale impact. Transit agencies need to operate their fleets reliably; respond quickly to failures in their operations; and adjust the bus routes on an ongoing basis based on changing travel patterns of passengers. Present day transportation management models utilize structured sensor data, or "telemetry," collected from buses including GPS location tracking of buses; number of passengers counted when entering or exiting buses; and whether buses follow their scheduled run times. While these types of telemetry collect *where* and *when* things happen, it does not explain *why* these events

happened. As the result of an event occurring while using public transportation (e.g., missing a stop), citizens report what occurred. For example, when a citizen reports, «автобус маршрут 48 не остановился на Байтурсынова» ("route #48 didn't make a stop at Baitursunov"), he/she provides valuable and actionable information about the nature of what occurred: route number; the stop that was skipped; and a geolocated position of the stop that was skipped (and therefore, if linked with city's geographic database system – GIS – could send a dispatch alert to a dispatcher in mere seconds).

The main reason behind this project was the need to automate the triaging process by using City Transportation Systems (CTS), which provides the public transportation service in Astana, Kazakhstan. CTS receives around 3 000 monthly complaints about the city's public transportation via three channels; a mobile app, a website portal and a call centre. All these complaints contain an open-ended description field (description) where customers report transportation problems with the city's public transportation system, including but not limited to missing bus stops, irregular routes, bad driver behavior and/or failed payments. At present, all complaints are processed by human operators who have to manually identify the relevant vehicle, stop and road associated with the complaint after reading it and then route the complaint to the applicable dispatcher. The manual processing of complaints is labor intensive; one single operator cannot process more than 60–80 complaints per hour, so when there is a large volume of complaints it may take up to 2-4 hours for them to reach the correct authority.

Automating the process of extracting the geotag from each complaint eliminates this "bottleneck". The triage/routing step is completely automatic if an NLP model is reliable to extract the Vehicle Identifier, Stop Name & Street Reference from each complaint. In addition, there are other substantial value-added analytical uses for the extracted geotags:

- **Identifying hotspots.** By analyzing the geographical grouping of complaints you can easily see which stops/intersections have higher than normal incident rates, thus allowing you to proactively maintain these areas as well as reschedule routes accordingly.

- **Analyzing trends at the route level.** By aggregating route numbers on a day-by-day basis you will be able to detect patterns of reliability degradation prior to when they become evident in your overall performance statistics.

- **Balancing dispatcher workloads.** By associating complaints with specific geographic zones (stops) you can better distribute the load among operators.

- **Overlaying GIS and visualizing data.** Once geotags have been successfully resolved against your City's stop/street database, you can create real-time situational awareness displays used by transportation management to help them visualize and understand current events.

- **Evaluating policy.** The before/after comparisons of the geotagged data associated with complaints can measure the effectiveness of the changes made to routes, investments in infrastructure or training programs for drivers

The nature of the CTS Complaint Corpus is difficult to handle due to its combination of different features that make it particularly challenging to be handled by conventional NLP pipelines. Firstly, these complaint texts contain code-switching - the switching from one language into another within or between sentences. Secondly, the informal tone used in these complaint texts contains informal language elements such as abbreviations (e.g., «м/а» = "route / bus"), spelling mistakes, and colloquialisms. In addition, there are large differences between the six geographic entity types, as well as their linguistic composition, occurrence rate and extraction difficulties; thus forming a heterogenous NER-task where no single modeling approach can be optimal for all entity-types at once. Finally, the data is very imbalanced on the classification level. The most frequent type (`STOP_NAME`) appears much more often (more than 10 times) than the least frequent type (`STREET_INTERSECTION`). Therefore a maximal likelihood-training will have many problems when trying to train the minority-class

The natural language processing (nlp) task most relevant to solving this problem is named Entity Recognition (NER). NER frames geotag extraction as a supervised sequence labeling problem in which each token within the input is assigned a semantic label that represents its class of entities and position within an entity span. The accuracy, robustness and computational efficiency of the chosen architecture for NER will directly determine whether or not it will be feasible to create an automated pipeline to recognize geotags for the CTS. This thesis develops and compares four fundamentally different architectures — fine-tuned multilingual transformer encoders; a Russian-specific BERT model; a multilingual embedding-based model; and large language models deployed via few-shot prompting — to identify which strategy or combination of strategies best meets the requirements of the operational needs of the CTS geotag extraction task.

The current research is based on an earlier study [1], which revealed the diverse characteristics of NER within transport domain, and demonstrated how aggregated evaluation metrics may hide some differences per category. In addition to this prior work,

the present thesis also develops a wide-ranging comparison of various models and offers concrete guidelines for the selection and deployment of NER models in the same type of environments as those mentioned above (low resource bilingual).

1.2 General characteristics of geographic named entities and toponyms in transport-domain texts

Geographic named entities (GNE's) that occur naturally in languages have many levels of abstraction: from large geographic units (countries, states/provinces, cities), to smaller ones (neighborhoods/districts, zip codes/postal zones), and then down to small, operational items (street names, building numbers, transit stop locations). In complaints about transportation issues, the GNE's most important for the operations are also very granular and specific to the domain; and they differ greatly from what is typically referred to as "spatial reference" in traditional NER benchmark models like CoNLL-2003, which groups all spatial references under one class called `LOC`.

Geotagged Complaints for Public Transport Operations Have Three Distinct Operational Uses *Complaint geotags are identification tags* (e.g. route number, vehicle identifier, license plate) that associate a complaint with an identifiable piece of equipment used in the field, which is stored in the fleet's management database. *Complaint geotags are also spatial tags* (e.g. name of stop at which incident occurred; name of road on which incident occurred), which allow a complaint to be associated with a particular location, such as one contained within the City's Geographic Information System (GIS) layers. Finally, *composite tags* (i.e., a combination of two or more spatial references combined into a single structured expression, e.g. "Intersection of Main St & Elm") provide even greater precision than spatial tags when referencing a particular location. As each type of geotag has its own unique structure, annotation characteristics and difficulties in extracting data

Due to corpus analysis and the domains' necessities; Six different types of entities were identified as necessary for the CTS geotag extraction task:

`ROUTE_NUM` Public transit route number. Examples: «Маршрут 47», «автобус № 64», «48 бағыт». These identifiers typically appear as short numeric spans, optionally preceded by a route keyword.

`BUS_ID` Vehicle identifier or fleet (garage) number assigned to a specific bus. Examples: «гаражный номер 3056», «борт 10815». Un-

like route numbers, bus IDs identify the physical vehicle rather than the service.

`PLATE_NUM` Vehicle licence plate number. Examples: «КЛС 012», «164 BW 01». Plates may combine Cyrillic and Latin characters in various regional format conventions.

`STOP_NAME` Named transit or bus stop. Examples: «аялдама Қобыланды», «остановка Сарыарка», «ТРЦ Керуен». Stop names are the most frequent entity type and appear in both Kazakh and Russian forms.

`STREET_NAME` Named street, avenue, boulevard, or road. Examples: «проспект Тауелсиздик», «улица Абая», «Кабанбай батыра даңғылы». Street names are morphologically inflected in context.

`STREET_INTERSECTION` Geographic intersection of two or more streets, expressed as a structured multi-word span. Examples: «угол Абая и Сейфуллина», «пересечение Тауелсиздик и Республики». This is the rarest and structurally most complex entity type.

Table 1.1 summarises the structural characteristics of the six entity categories, including their linguistic nature, typical span length in tokens, and the principal sources of extraction difficulty.

Table 1.1: Structural characteristics of entity categories in the CTS corpus (adapted from [1]).

Entity Type	**Nature**	**Typical Span (tokens)**	**Principal Extraction Difficulty**
`ROUTE_NUM`	Numeric / identifier	1–4	Boundary ambiguity with time codes and QR identifiers
`BUS_ID`	Numeric / identifier	1–3	Semantic confusion with `PLATE_NUM`
`PLATE_NUM`	Alphanumeric	1–3	Mixed Latin-Cyrillic format variation
`STOP_NAME`	Nominal / toponym	1–8	Abbreviation, bilingual lexical variation
`STREET_NAME`	Morphological	2–6	Case inflection, code-switching boundary
`STREET_INTERSECTION`	Composite / structured	5–12	Internal compositional structure; rare training examples

`ROUTE_NUM` and `STOP_NAME` appear to be the most frequent entity types as they can be found on about 72 % and 68 % of complaints, respectively. The rarest type is `STREET_INTERSECTION` which appears in less than 3 % of cases. Structurally, `STREET_INTERSECTION` is the most complicated as it has a span of 5–12 tokens including conjunction words (*угол*, *пересечение*, *қиылысы*) with both distinct street name forms with grammatical case marker, and optionally contains additional information such as direction or time.

Transportation domain toponyms have many differences to general language toponyms which make it difficult to do the automatic toponym extraction. *Context dependence:* The same word e.g. "Republic" can be used as a reference point (e.g. bus stop name, street name, organization) based on contextual references. *Abbreviation:* Stop names and street types are often shortened when written in complaint texts (e.g. «ул.» for «улица», «пр.» for «проспект», «м/а» for «маршрут/автобус»). *Orthographical Variants:* Both Russian and Kazakh languages use different forms of writing the same stop name; e.g. «Байтурсынова» (Russian) and «Байтурсынов даңғылы» (Kazakh) within the same document collection and thus require a model to generalize across those

variations. *Identifier Overlap:* BUS_ID and PLATE_NUM both look like the same type of token, i.e. small number of characters alphanumeric strings but have limited contextual support, resulting in high overlap rates in models that primarily depend on surface form.

Token distributions are very unbalanced. Analysis of dataset shows that the `O` (Outside) class comprises about 85 % of the total tokens, while entity tokens represent fewer than 14 %. Between these two entity token types, `STOP_NAME` and `ROUTE_NUM` are present in around 66 % of all entity instances while `STREET_INTERSECTION`has fewer than one percent. As such, strong class weighting methods in training objectives, data augmentation methods through sampling techniques, and macro-average evaluation metrics which will assign equal weight to each entity category regardless of its occurrence rate are required.

1.3 Modern approaches to automatic named entity recognition

Named entity recognition (NER) has been defined by three main paradigms in the last thirty years, with each paradigm offering different balances between coverage of language, the requirement for labelled training data, processing costs, and NER performance under low resource or code switching conditions.

1.3.1 Rule-based and statistical methods

The first NER systems were built solely with hand-crafted rule sets. A dictionary of recognized named entities (gazetteer) along with a collection of regular expressions to match standardized writing formats for named entities (such as capitalization, punctuation and number sequences), along with manual coding of contextual knowledge based on these structures formed the base for systems such as ANNIE and LingPipe. While regex based methods can be very accurate for structurally defined identifiers that have been standardized in some way (e.g., license plates; route numbers; postal codes), the method has low recall rates when it does not include all possible variants within its definition.

Hidden Markov Models (HMM) extended Rule-Based Systems with a Probabilistic Sequence Model. In an HMM based NER System, the observed Output is the Token

Sequence and the Hidden States are the BIO Labels. The models jointly maximize:

$$P(\mathbf{y}, \mathbf{x}) = \prod_{i=1}^{n} P(y_i \mid y_{i-1}) \cdot P(x_i \mid y_i) \tag{1.1}$$

Where $P(y_i \mid y_{i-1})$ represents Transition Probability from one label to another Label, and $P(x_i \mid y_i)$ is the Emission Probability for Token x_i as it relates to label y_i. These optimal Label Sequences can be Decoded by using the Viterbi Algorithm. An HMM does model First Order Dependencies in Labels but will not allow Long Range Contextualization nor arbitrary Feature Functions.

Conditional Random Fields (CRF) [2] provided an alternative to HMMs as they modeled the *conditional* probability $P(\mathbf{y} \mid \mathbf{x})$, in place of modeling the joint. Linear chain CRF are defined as follows:

$$P(\mathbf{y} \mid \mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp\left(\sum_{i=1}^{n} \sum_{k} \lambda_k f_k(y_{i-1}, y_i, \mathbf{x}, i) \right) \tag{1.2}$$

where f_k denote feature functions, λ_k represent learned weights, and $Z(\mathbf{x})$ denotes the normalization factor. Feature functions in linear chain CRFs may be used to define arbitrary features from input sequences (e.g., prefix/suffix patterns, capitalization, membership in dictionaries). Richly engineered CRFs were able to hold top NER performance for almost a decade.

BiLSTM-CRF represents the connection between traditional Statistical Methods and New Neural Models. Bidirectional LSTMs take advantage of both left-to-right and right-to-left contextual information when encoding each token as follows:

$$\overrightarrow{\mathbf{h}}_i = \mathrm{LSTM}_{\mathrm{fwd}}(e_i, \overrightarrow{\mathbf{h}}_{i-1}), \quad \overleftarrow{\mathbf{h}}_i = \mathrm{LSTM}_{\mathrm{bwd}}(e_i, \overleftarrow{\mathbf{h}}_{i+1}) \tag{1.3}$$

$$\mathbf{h}_i = [\overrightarrow{\mathbf{h}}_i; \overleftarrow{\mathbf{h}}_i] \tag{1.4}$$

Both hidden states are concatenated into one single hidden state $\mathbf{h}_i$. This hidden state is fed into a linear layer that produces the label logit scores. The logit scores are then decoded collectively over all tokens by the CRF layer. The CRF ensures globally consistent sequences of labels (for example, an "I-" tag will always follow a "B-" tag of the same part-of-speech category). Prior to being surpassed by Transformer models, BiLSTM-CRF was able to achieve the highest score in the named-entity recognition task on CoNLL-2003; it is still used today as a strong competitor to many other models, especially those that have limited training data.

Rule-based and statistical models lack generalizability to morphologically complex languages or low resource languages due to their dependency on manually engineered features.

1.3.2 Transformer-based encoders (BERT, RoBERTa, XLM-R)

The transformer architecture [3] included the self-attention model, which computes context representation by considering the entire input at once through simultaneous position-wise attention. Therefore for an arbitrary input:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1.5}$$

there are Query Matrix Q, Key Matrix K and Value Matrix V. We also know that the Key Dimension d_k. Each Head (H) in multi head attention is applied separately and then the outputs are concatenated together as follows:

$$\text{MultiHead}(Q, K, V) = [h_1; h_2; \ldots; h_H]W^O, \quad h_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{1.6}$$

The method allows a single token to pay attention to every other token within the sequence regardless of position; it eliminates the vanishing gradient problem that had limited how long recurrent networks could process sequences.

BERT [4], was based upon the concept of the "pre-training" followed by "fine-tuning" paradigm used for NLP. In its "pre-training", there are two unsupervised training objectives applied to large amounts of unlabelled data. The first is Masked Language Modeling (MLM), where 15 % of all input tokens are randomly masked; this causes the model to be trained on predicting the actual token from the context.

$$\mathcal{L}_{\text{MLM}} = -\sum_{i \in \mathcal{M}} \log P(x_i \mid \mathbf{x}_{\setminus \mathcal{M}}) \tag{1.7}$$

The model is trained using a variant of the Masked Language Modelling objective. Where $\mathcal{M}$ is the set of masked positions. Next Sentence Prediction (NSP) objectives trains the model to predict whether two sentences (sentence segments) can occur consecutively in the original document. After pre-training, a small task specific-head (a single linear layer for token classification) is added to the top of the model and then the entire model is fine-tuned on labeled data from the task. Bert-base achieves an f-score of 91.3 on CoNLL-2003 and significantly exceeds baseline performance based on a bi-lstm crf.

RoBERTa improved over Bert through removal of the next sentence prediction objectives (this was later found to be detrimental to future models), increased training batch size (from 256 to 8000), extended the amount of training time (bert used 16gb of text while robertabase used 160gb of text) and a dynamic masking method that produced new mask patterns each epoch. This provided better quality and less variability in representation; thus, robertalarge achieved an f-score of .93 on CoNLL-2003 [5].

XLM-RoBERTa (XLM-R) [6], expanded roberata's pre-train process into 100 languages, utilizing 2.5 tb of common crawl text after filtering. The shared subword vocabulary of 250 000, as defined by Sentencepiece, allowed xlm-r to leverage cross lingual knowledge transfer; thus similar semantic phrases across different languages clustered in their respective embedding spaces. Xlm-r base uses 12 layers of transformers, 768 hidden dimensions, 12 attention heads and approximately 125m parameters. Its ability to support both Kazakh and Russian in a singular vocabulary, along with its other properties make it highly suitable for the code switching nature of the cts dataset. A formal evaluation of xlm-r relative to m-bert has shown that xlm-r performs significantly better than m-bert at NER on all 16 test languages, and also performs relatively well on morphologically complex Slavic and Turkic languages. [6].

The XLM-R encoder for the NER task is enhanced using an additional token classification head: The classification head consists of a single linear transformation $W \in \mathbb{R}^{768\times K}$ maps each contextualized word representation $\mathbf{h}_i \in \mathbb{R}^{768}$ into a vector of class logits. Then, after applying a softmax activation function, we obtain the probability distribution over all possible labels:

$$p(y_i = c \mid \mathbf{x}) = \frac{\exp(\mathbf{h}_i^\top \mathbf{w}_c)}{\sum_{j=1}^{K} \exp(\mathbf{h}_i^\top \mathbf{w}_j)} \tag{1.8}$$

Here $K = 13$ corresponds to the total amount of different labels in our dataset (i.e., 6 entity types $\times$ 2 BIO prefixes $+ O$).

1.3.3 Embedding-based pipelines (E5, multilingual models)

A parallel body of work generates dense text embeddings optimized to model semantic similarity and support information retrieval tasks via either an instruction-tuned or contrastive learning paradigm as opposed to a masked language modeling paradigm. For example, these models generate representations where semantically similar texts are similarly situated within the representation space regardless of their superficial characteristics.

Multilingual-E5 (mE5) [7] represents a set of models based upon the XLM-RoBERTa architecture. mE5 was trained on approximately one billion multilingual text pair examples utilizing a multi-stage process that first utilized weak supervision from noisy parallel data followed by fine tuning against high quality supervising data derived from MS MARCO, Natural Questions and TriviaQA. The primary objective of the training function is to provide a contrastive loss that will pull together the representation(s) of each pair of semantically equivalent text while pushing apart the representation(s) of non-equivalent pairs:

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau)}{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau) + \sum_j \exp(\text{sim}(\mathbf{q}, \mathbf{d}_j^-)/\tau)} \tag{1.9}$$

where $\mathbf{q}$ is a query embedding, $\mathbf{d}^+$ is a positive document, $\mathbf{d}_j^-$ are in-batch negatives, and τ is a temperature parameter.

While originally intended for use at the sentence level, mE5's contextual token representations may be used at the token level using a token based projection instead of pooling tokens together by their sentences. For the purposes of this thesis the `multilingual-e5-base` (278M parameters) was chosen because it struck a balance between high-quality vector representations and reasonably good computational performance.

Petasis and Karkaletsis [8] show that E5-Large Vector Representations combined with an ensemble of traditional machine learning classifiers (XGBoost, k-Nearest Neighbor, Logistic Regression), through Soft Voting, achieves Accuracy of .92 and F1 of .91 on a low resource Greek Named Entity Recognition Task, substantially better than several state-of-the-art Transformer Fine-Tuning Baselines. These results indicate that strong pre-trained vector representations can be sufficient to overcome limitations associated with small amounts of labeled training data, which will likely be true for many of the rare named entities found in the CTS Corpus.

1.3.4 Large language models for few-shot NER

Recent years have seen the rise of instruction-tuned large language models (LLMs)—models with hundreds of billions of parameters that were trained on vast multi-task data sources and enable task adaptation through natural language prompts instead of gradient-based weight updates. The few-shot paradigm [9] provides the model with a small number of task-specific examples in the prompt and asks it to generalize to new inputs without any weight updates.

In-context learning for NER frames the extraction task as a conditional text generation problem: given the system prompt, few-shot examples, and the input complaint text, the model generates a JSON object mapping entity types to extracted spans. This approach does not require labelled training data beyond the couple of shots used as examples, which makes it very attractive for entity type with very limited training instances.

Qwen2.5 [10], is an open-wight llm series (0.5b – 72b parameter count), developed by Alibaba. The 7B-parameter variant supports 29 + languages including Russian and Kazakh, and showcase strong structured output generation using instruction tuning on supervised fine-tuning (SFT) and Reinforcement Learning from human feedback (rlhf) datasets. Instruction tuning is crucial for the JSON structure of the NER output: a model must not only identify entity spans but also reliably format the output as valid JSON with correct field names.

Performance comparison of different LLMs in terms of NER performance. Keraghel et al. [5] report that GPT-4 in zero-shot mode has a value of F1=0.83 on conll-2003. This value reduces to within 10 percentage points of finetuned RoBERTa-large (F1=0.93). Llama3.1 and mistral have lower values of performance (F1$\approx$0.60–0.70) on this benchmark, especially on entities with limited lexical cues. four LLMs are evaluated under identical few-shot conditions in this thesis to distill the effect of capacity and multilingual pretraining coverage.

The design of prompts for the NER task using an LLM will influence how well it can perform. The entity types included in the prompt define what entities the system should look for in addition to defining extraction guidelines for those entities as natural language. Examples provided to the system were chosen so that all entity types would be represented and specifically, so that the least frequent classes (`STREET_INTERSECTION`, `STREET_NAME`) would be represented because it is likely these types would not have been encountered by the model during pre-training enough times to have developed knowledge about them.

1.4 Using machine learning and natural language processing for multilingual text analysis

Multimodal NLP seeks to develop Language Technology for multiple languages simultaneously; especially for those who do not have a large amount of data available in

standard benchmark datasets. The Kazakh-Russia bilingual environment poses its own set of unique challenges for developing the system architecture.

Code-switching occurs when there is intra-sentential mixing of two or more languages per single utterance. Approximately 67 % of complaint records within the CTS dataset include at least one sentence-internal language shift. An example is as follows: «м/а 48, уақыты 09:30, остановка Тауелсиздик бағытында жоқ» (route 48, time 09:30, no stop in the direction of Tauelsizidik), where transit terminology in Russian is combined with temporal and directional phrases in Kazakh. Because code-switching violates the monolingual assumptions inherent to single language training corpora, it can result in out-of-vocabulary errors or diminished representations for tokens in the less represented language.

Kazakh morphological complexity. Kazakh is an agglutinative language in which grammatical information (such as case, number, possession, tense) is expressed using a string of affixes attached to the root of the word. One noun can exhibit up to nine different surface forms depending upon the grammatical case being used. Therefore, if a geotag is required to recognize a "stop" named «Байтурсынов», then it could occur in any of the following forms: «Байтурсыновта» (locative: at Baitursunov); «Байтурсыновқа» (dative: toward Baitursunov); or «Байтурсыновтан» (ablative: from Baitursunov). Although sub-word tokenization (using Sentence Piece BPE) does help with representing inflectionally complex words as composed of root and suffix pieces, the BIO tags still need to be assigned based upon whole word units to ensure annotation consistency.

Russian morphological complexity. As a fusional language, Russian has six cases and two genders; gender affects how adjectives agree with nouns. Streets in Russia are named using varying cases based on the syntactical position they occupy: «проспект Абая» (nominative) versus «на проспекте Абая» (prepositional) versus «по проспекту Абая». Regardless of whether there is a case-inflectioned preposition or not, each entity includes all of the full canonical parts of the street name.

Low-resource asymmetry. Kazakh is low resource relative to Russian within most multilingual pre-training corpora. For example, while Russian is among the top ten most represented languages in CommonCrawl, the representation of Kazakh is much smaller. The result of this difference is that multilingual models will likely have less capacity to represent morphological features specific to Kazakh resulting in suboptimal subword representations when representing Kazakh tokens that co-exist with other languages within an input.

Cross-lingual transfer learning refers to how a multilingual model uses information it

has learned about one language in order to help with the improvement of its performance when using another language. A key part of this is that there is an implicit alignment across languages within the shared subword vocabulary of XLMRoBERTA between the representations of semantically-related token types. The representation of «маршрут» (Russian: route) will be similar to «бағыт» (Kazakh: route) even though they have no common surface form. This cross-language alignment is also important for CTS since the same entity type could be expressed in either language

Iqbal et al. [11] demonstrated that multilingual BERT (mBERT) achieved a macro F1 score of 0.98 on an augmented multilingual NER dataset and exceeded the monolingual RoBERTa (F1 = 0.884). Iqbal et al. attribute mBERT's superiority over monolingual RoBERTa to cross-lingual transfer rather than solely due to increased model capacity, thereby confirming that multilingual pre-training provides benefit even for individual-language NER tasks. Kassab et al. [12] found that DeepPavlov RuBERT remains the strongest single Russian-language NER model, achieving a 28 % improvement in precision over commercial competitors due to its pre-training specifically on large amounts of Russian web content. RuBERT can still be applied to Kazakh-Russian code-switching corpora, but only with substantial adaptation: a class-weighted cross-entropy training objective is required to prevent the loss from collapsing on minority entity types, an extended training schedule (10 epochs with linear warmup and early stopping on validation F1) is needed for the model to converge, and a careful subword-to-character label alignment is necessary to compensate for the WordPiece tokeniser's poor coverage of Kazakh morphemes. Under these conditions, RuBERT reaches moderate but uneven performance on the CTS task, with strong recognition of the lexically regular `ROUTE_NUM` class and considerably weaker results on entity types that depend on Kazakh-language context. This reflects the fundamental limitation that the underlying WordPiece vocabulary contains no dedicated tokens for Kazakh inflectional morphology, irrespective of the downstream training procedure.

Tao et al. [13] show that the use of BERT + BiLSTM + CRF and NLP specific preprocessing achieve an F1 score of 0.93 on a Chinese language dataset, with geographical annotations from social media. They also demonstrated that using contextual embeddings together with preprocessing for informal registers can significantly increase the accuracy of predictions based on noisy, user-generated content. The findings by Tao et al. are directly applicable to the CTS corpus because it has several similarities to social media text (informal register, abbreviations, inconsistent punctuation, etc.) as well as embedded entity references within unstructured prose.

Geoparsing — combining geographic entities into their respective latitude/longitude coordinates through both geographic entity recognition (GER) and coordinate resolution—is an extension of named-entity recognition (NER). Hu et al. [14], assessed 27 geoparsers from 26 publicly available data sets. The authors reported that BERT-type models achieved the best F1 scores (of all the evaluated models) for the task of identifying geographic names in social media texts (i.e., TopoBERT F1 = 0.854), while the performance of traditional geoparsers (e.g., Stanford NER) fell off significantly when tested against very noisy inputs. Moncla et al. [15], developed methods using CamemBERT and XLM-RoBERTa to improve upon prior work by resolving ambiguities associated with topographic terms; these improvements were demonstrated on multiple French geographic data collections. The authors further reported that the use of neural classifiers on top of the embedding space provided by transformer architectures could resolve spatial ambiguity issues where other systems failed due to reliance on rules.

Sun et al. [16], illustrated that incremental training of BERT (on a domain-specific corpus) to support a Chinese social media language environment, when followed by a combination of conditional random field (CRF) and error correction processing (ChineseCTRE), produced a significant increase in the recall of geographic NER (4-5 percent greater than baseline BERT), and therefore indicated that domain-adaptive pre-training of large-scale pre-trained models can be an effective method for improving the performance of minority classes within specific domains.

1.5 Comparison of NER architectures for low-resource and code-switched (Russian–Kazakh) settings

Table 1.2 summarises the 21 key related works surveyed for this thesis, spanning rule-based, statistical, encoder, embedding, and LLM-based NER approaches. The works are presented in the order introduced in the thesis presentation.

Table 1.2: Summary of key related works on NER and geotag extraction.

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
1	Yang et al. (2023) [9]	BERT, GPT-2/3/4, T5, LLaMA, BLOOM	Survey (no dataset)	Covers evolution from BERT to ChatGPT across pretraining objectives and NLP tasks

Continued on next page

Table 1.2 continued from previous page

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
2	Keraghel et al. (2024) [5]	BiLSTM-CRF, BERT, RoBERTa, GPT-3/4, LLaMA	RoBERTa F1 = 0.93; GPT-4 zero-shot = 0.83	Fine-tuned encoders dominate supervised NER; LLMs excel in few-shot settings
3	Hu et al. (2023) [14]	27 tools incl. TopoBERT, spaCy, Stanford NER	TopoBERT F1 = 0.854	BERT-based models best on social media; classical tools degrade sharply on noisy text
4	Iqbal et al. (2024) [11]	mBERT, RoBERTa-small/large, BERT-base/large	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on augmented multilingual NER
5	Kassab et al. (2025) [12]	DeepPavlov RuBERT, Natasha, Stanza, spaCy	+28% precision over single RuBERT	Semi-automated multi-model voting reduces manual annotation effort drastically
6	Petasis & Karkaletsis (2024) [8]	E5-large + XGBoost, KNN, LogReg (soft voting)	Accuracy = 0.92; F1 = 0.91	E5 embeddings + ML ensemble beats fine-tuned transformers on low-resource NER
7	Wang et al. (2023) [7]	mE5-small/base/large/large-instruct	BEIR score = 64.4 (large-instruct)	Two-stage contrastive training yields strong multilingual retrieval embeddings
8	Yang et al. (2024) [10]	Qwen2.5 (0.5B–72B, Base, Instruct, MoE)	MMLU, HumanEval benchmarks	Instruction tuning improves structured JSON output; open-weight models cost-effective
9	Mai et al. (2024) [17]	Mistral, Baichuan2, Llama 2, Falcon + geo-KB	Improved geoparsing precision/fairness	LLM fused with geographic knowledge bases improves geoparsing for underrepresented regions
10	Tarannum et al. (2026) [18]	mBERT, RoBERTa-small, BERT-large/base	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on heterogeneous address data
11	Wei et al. (2024) [19]	ALBERT-Conv1D-BiLSTM-CRF	P = 0.981; R = 0.967; F1 = 0.974	Fusing global (ALBERT) and local (Conv1D) features improves historical toponym NER
12	Moncla et al. (2022) [15]	CamemBERT, mBERT, XLM-RoBERTa + classifier	CamemBERT F1 ≈ 0.92	Neural classifiers on contextual transformer embeddings resolve toponym ambiguity
13	Tao et al. (2022) [13]	BERT + BiLSTM + CRF + NLP preprocessing	F1 = 0.93	Enhanced BERT with preprocessing excels on noisy social media geographic NER

Continued on next page

Table 1.2 continued from previous page

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
14	Manjavacas et al. (2026) [20]	GPT-4o, GPT-OSS-120B, Llama-3.1-8B	GPT-4o ≈100%; Llama ≈73%	Larger models handle complex narrative NER fields better than smaller open-weight models
15	Stankevičius & Lukoševičius (2024) [21]	BERT-110M, T5, Random Embeddings	Repr.-shaping ≈0.70 vs. baseline ≈0.50	Simple representation shaping can match complex BERT models on similarity benchmarks
16	Mansourian & Oucheikh (2024) [22]	Llama 2 (LangChain + RAG) + NLP extraction	ChatGeoAI ≈25% on complex tasks	LLMs need RAG and entity mapping to execute complex geospatial workflow tasks
17	Lutsai & Lampert (2024) [23]	Multilingual BERT + multitask PMOP regression	PMOP: 96% acc.; 3.68 km median error	Custom multitask loss with PMOP achieves lowest spatial errors in tweet geolocation
18	Alsaqer et al. (2023) [24]	LR, SVM, RF, Naïve Bayes	Accuracy = 67%	Classical ML predicts tweet location from text; accuracy limited by short noisy text
19	Santos et al. (2021) [25]	BERT-based vs. HMM/CRF baselines	BERT F1 ≈ 92–95%; HMM/CRF ≈ 80–85%	BERT significantly outperforms HMM/CRF for parsing unstructured real-world address text
20	Sun et al. (2023) [16]	ChineseCTRE (BERT + BiLSTM-CRF, incr. pre-training)	F1 ≈ 91–92% vs. baseline 87–88%	Incremental domain pre-training with error correction boosts geographic NER recall
21	Ussenbayeva et al. (2026) [1]	XLM-RoBERTa (base), Regex, CRF, Frozen LM	ROUTE_NUM F1 = 0.920; STREET_NAME F1 = 0.282	Transport NER is heterogeneous; aggregated metrics conceal per-class disparities; composite entities require specialised modelling

Synthesis of literature findings. A number of common patterns emerged from the literature review that have a significant impact on the design decisions made in this thesis.

Transformer encoders dominate supervised NER. In almost all cases where sufficient labeled training data are available, fine-tuning pre-trained transformer based encoders (bert, roberta, XLM-roberta) significantly outperforms both traditional rule-based named entity recognition systems as well as classical statistical models (crf, BiLSTM-CRF). This performance difference is especially dramatic on noisy informal content such as social media or complaints, in which hand-crafted features and gazetteers fail to generalize.

Multilinguality is non-negotiable for code-switching corpora. Works 4, 10 and 12 collectively demonstrate that using multilingual models (mbert, XLM-roberta), in addition to being necessary for most real-world applications, provide substantial improvements over their monolingual counterparts when there is mixed language content in the evaluation corpus; regardless of whether the monolingual model was specifically trained for each of the target languages.

Competitive Learning Models for Rare Entity Types. The results in works 1, 2, 8, 9 and 14 demonstrate the instruction-tuned learning models' ability to be most suitable for the extraction of rare entity types. Those entity types typically have fewer examples than other types used during training of fine-tuned encoders. Therefore, there will be overfitting and/or label collapse issues with those fine-tuned encoders. For example, in the CTS dataset, the entity type `STREET_INTERSECTION` is only trained by two examples. Thus, LLM's few-shot extraction abilities are the only possible means to extract `STREET_INTERSECTION` from the given data.

Aggregate metrics conceal category-level heterogeneity. The authors note in their paper 21 (the authors previous research) that the use of a single average F-score can give an overly optimistic view of performance when using a multi-type entity taxonomy. Since the operations costs associated with incorrectly classifying a `STOP_NAME`versus not extracting a `STREET_INTERSECTION` are different, per-type evaluation is required to accurately evaluate the systems performance and determine if they should be deployed into production.

Gap analysis. To date no one has developed a system that addresses all of the following: (a) uses code switching languages (Kazakh & Russian), (b) extracts information from text related to transportation complaints, (c) utilizes a six-type entity classification model including composite space-based entities, and (d) compares four architecturally different methods within the same experimental framework. The authors identify this gap as the basis of the contribution of this thesis.

1.6 Setting the task

1.6.1 Formulation of the purpose and objectives of the thesis on automatic geotag recognition

The research aim of this thesis is to develop and evaluate a range of machine learning techniques for the automatic recognition of geographic references in multilingual text,

where the text consists of public transportation complaints written in a combination of Russian and Kazakh, and to assess the accuracy and robustness of each technique across structurally heterogeneous entity types. The research object is the process of identifying geographic named entities within multilingual texts related to public transportation complaints collected by City Transportation Systems (CTS) of Astana, Kazakhstan. The research subject consists of the family of machine learning models and algorithms for the automatic extraction of six different types of geotag entities from noisy, code-switched Russian–Kazakh natural language text.

To accomplish this aim the thesis pursues six interlinked objectives. The first objective is to perform an exhaustive review of the techniques currently used for automatic geotag detection and named entity recognition, with particular emphasis on those approaches that operate effectively in multilingual and low-resource environments. The second objective is to gather, preprocess, and manually annotate a domain-specific multilingual dataset of public transportation complaint records, producing gold-standard BIO labels for six entity types. The third objective is to design and implement four distinct machine learning strategies for geotag recognition: a fine-tuned XLM-RoBERTa encoder, a fine-tuned DeepPavlov ruBERT model adapted to the code-switching corpus through a class-weighted training objective and an extended training schedule, a fine-tuned Multilingual-E5 embedding-based model, and a few-shot prompting pipeline based on the Qwen2.5 large language model. The fourth objective is to implement four baseline methods — a regex rule extractor, a frozen XLM-R linear probe, a handcrafted-feature CRF, and a most-frequent-class predictor — in order to place the performance of the four main approaches in context. The fifth objective is to evaluate and compare all models using per-class precision, recall, and F1-score, supplemented by confusion matrix analysis and boundary stability analysis. The sixth and final objective is to determine which modelling strategy is best suited to each entity type, and which overall strategy is most appropriate for real-time operational deployment within the CTS complaint-handling workflow.

1.6.2 Requirements for an intelligent geotag-extraction system

The intended deployment context, namely the real-time triage of CTS complaint records, imposes a set of functional and non-functional requirements on the geotag extraction system. The system must process Kazakh, Russian, and code-switched sentences directly, without requiring an explicit language identification step as a precondition for analysis. It must classify extracted spans into the six domain-specific entity types defined earlier

in this chapter, namely `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`. The system must achieve span-level precision of at least 0.70, because the downstream geocoding workflow is highly sensitive to false positives and high precision is the primary operational constraint that prevents the generation of spurious dispatch alerts. The recall on the two most frequent and operationally critical entities, `STOP_NAME` and `ROUTE_NUM`, must reach at least 0.80, since these classes form the primary input to dispatch routing and to geographic hotspot detection.

In addition to the per-class quality requirements, the system must satisfy non-functional constraints that arise from the operational environment. It must achieve sufficient computational throughput to be deployed in real time on standard server hardware, processing at least one hundred records per minute, which in practice places a limit on the size of the model and on the latency of each inference call. The system must also degrade gracefully on noisy input, tolerating spelling errors, missing punctuation, non-standard abbreviations, and mixed-script segments that are typical of citizen complaint text. Finally, the extracted entities must be returned as structured data with explicit entity types, exact text spans, and character offsets, so that the output can be consumed directly by the downstream GIS linking and database insertion components.

1.6.3 Formulation of data requirements for model training

The task can be formally defined as a token-level sequence labelling problem under the BIO (Beginning-Inside-Outside) tagging scheme. Given an input token sequence $\mathbf{T} = (t_1, t_2, \ldots, t_n)$, the objective is to assign each token t_i a label $l_i \in \mathcal{L}$, where the label set is defined as

$$\mathcal{L} = \{B\text{-}e,\ I\text{-}e \mid e \in \mathcal{E}\} \cup \{O\}, \quad |\mathcal{L}| = 2|\mathcal{E}| + 1 = 13 \tag{1.10}$$

and $\mathcal{E} = \{\texttt{ROUTE_NUM}, \texttt{BUS_ID}, \texttt{PLATE_NUM}, \texttt{STOP_NAME}, \texttt{STREET_NAME}, \texttt{STREET_INTERSECTION}\}$ denotes the set of the six entity types, with O denoting the outside (non-entity) class. The BIO scheme encodes entity spans such that the first token of an entity receives the B- prefix, all subsequent tokens within the same span receive the I- prefix, and all non-entity tokens are labelled O. A predicted entity span $\hat{s}$ is counted as a true positive only when both its left and right boundaries exactly match the gold annotation and its entity type label is correct. The scheme also imposes structural constraints on valid sequences: an I-e label must be immediately preceded by either B-e or I-e of the same entity type e, a B-e label may follow any label including an I-e' of a different type

$e' \neq e$ (which simultaneously terminates the previous entity and begins a new one), and an O label always terminates any open entity span.

Several requirements on the data follow directly from the formulation. The annotated corpus must contain at least two thousand labelled complaint records in order to support fine-tuning of transformer encoders without severe overfitting. The training and test splits must be produced through stratified sampling that preserves the proportion of each entity type, so that the test distribution remains representative of the full corpus and per-class evaluation remains meaningful. The annotations themselves must be produced as a gold standard by at least two independent annotators with native or near-native proficiency in both Kazakh and Russian, with all disagreements resolved through adjudication. The corpus must also provide adequate coverage of the rarest entity type, with a minimum of fifty training instances of `STREET_INTERSECTION`, since fine-tuning a transformer model on a class with fewer instances becomes statistically unreliable. The data should be stored in JSONL format with pre-tokenised BIO label sequences so that the training pipeline can use efficient DataLoader-based batching without performing tokenisation on the fly.

The severe skew of the entity distribution discussed in Section 1.2 introduces an additional methodological requirement: the training procedure must include explicit strategies to prevent the loss from collapsing onto majority classes. The principal strategy is a weighted cross-entropy loss in which the weight of class c is computed as $w_c = N_{\text{total}}/(K \cdot N_c)$, where N_c is the count of class c and K is the number of classes; this construction gives minority entity classes proportionally higher influence on the gradient. This is complemented by oversampling, in which training records that contain rare entity types are duplicated several times within the training pool, increasing their effective representation without requiring additional manual annotation. Finally, macro-averaged evaluation is adopted as the primary reporting metric, so that strong performance on frequent classes cannot mask failures on rare ones.

1.6.4 Defining criteria for evaluating the effectiveness of NER models

Model performance is assessed using standard span-level NER metrics computed by the `seqeval` library, which requires exact boundary and label matching for a true positive. A predicted span $\hat{s} = (\hat{t}, \hat{l}, \hat{r})$ described by its type, left boundary, and right boundary is a true positive if and only if there exists a gold span $s^* = (t^*, l^*, r^*)$ such that $\hat{t} = t^*$,

$\hat{l} = l^*$, and $\hat{r} = r^*$ all hold simultaneously. The per-class precision, recall, and F1-score follow directly from the counts of true positives, false positives, and false negatives for each class:

$$\mathrm{P}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \quad \mathrm{R}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \quad F_{1,c} = 2 \cdot \frac{\mathrm{P}_c \cdot \mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c} \tag{1.11}$$

The macro-averaged F1 weights all entity types equally regardless of their frequency in the corpus and is computed as

$$F_{1,\mathrm{macro}} = \frac{1}{|\mathcal{E}|} \sum_{c \in \mathcal{E}} F_{1,c} \tag{1.12}$$

and is adopted as the primary evaluation metric because of its sensitivity to performance on rare classes such as `STREET_INTERSECTION` and `STREET_NAME`. By contrast, the micro-averaged F1 aggregates true-positive, false-positive, and false-negative counts across all classes before computing the ratio,

$$F_{1,\mathrm{micro}} = 2 \cdot \frac{\sum_c \mathrm{TP}_c}{\sum_c (2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c)} \tag{1.13}$$

and is therefore influenced primarily by the most frequent classes; it provides an overall measure of throughput-weighted performance that is more appropriate for system-level reporting than for per-class diagnostics.

A complementary diagnostic measure is the boundary stability for each class, which captures the fraction of predictions that are correctly typed and also correctly delimited:

$$\mathrm{BS}_c = \frac{|\{(t, l, r) \in \hat{S}_c : \exists\, (t^*, l^*, r^*) \in S_c^*,\ l = l^* \wedge r = r^*\}|}{|\hat{S}_c \cap S_c^*|_{\text{type-only}}} \tag{1.14}$$

where $|\hat{S}_c \cap S_c^*|_{\text{type-only}}$ counts the spans that match in entity type but not necessarily in boundary. A model that exhibits high precision together with low boundary stability is one that correctly identifies what an entity is but tends to over- or under-extend its boundary, a behaviour that introduces silent errors into downstream geocoding. Beyond the numeric metrics, a confusion matrix at the entity-type level (collapsing BIO prefixes) is used to identify systematic misclassification patterns. For the CTS taxonomy the most informative confusion axes are between `BUS_ID` and `PLATE_NUM`, since both classes are realised as short alphanumeric identifiers, and between `STOP_NAME` and `STREET_NAME`, since both are multi-word spatial toponyms that frequently share surface forms; confusion concentrated along either of these axes indicates that the model has relied on surface form rather than functional context.

Taken together, the per-class scores, macro- and micro-averaged F1, boundary stability, and the entity-level confusion matrix constitute an operationally motivated evaluation framework that moves beyond any single aggregate score and supports category-aware model selection for deployment.

Chapter 2

Model development and experimental setup

The full process of the geotag recognition task's experimental workflow is presented in this chapter. In Section 2.1, data acquisition, as well as annotation, preprocessing, and partitioning are described. Section 2.2 provides an overview of all four machine learning methods that are being evaluated (the reasons behind each architecture and what the training objective was), and in Section 2.3 a description of the reference baseline method is provided. Finally, in Section 2.4, the evaluation methodology is formalized.

2.1 Data collection and preparation

2.1.1 Data collection: transport complaint records from City Transportation Systems

The main data collection point is the Operational Complaint Database maintained by the City Public Transport Systems (CTS) - the operator of Astana's public transport system. Each month, it receives around 3 000 complaints via three input channels; a passenger-facing mobile app, an official city portal and a call centre that transcribes calls into complaints. Every single complaint has to be entered with a mandatory free-text description field (*описание*) where the customers have described incidents involving public transport. These include missed bus or tram stops, deviation from planned routes, drivers' behavior and issues related to fare payments as well as conditions of vehicles. The total number of complaints made during this twelve months of operation time was then anonymized by CTS before being handed over to us. All personal identification details had been deleted at that point. There are no geographic coordinates available in

relation to each individual complaint.

The raw data contains several elements which distinguish it from other benchmarks used in NLP. Firstly, within approximately 67 % of entries there is evidence of intra-sentential switching between Kazakh and Russian. For example «маршрут 48, уақыты 09:30, остановка Тауелсиздік бағытында жоқ» (route 48, time 09:30, stop in the direction of Tauelsizidik absent). Secondly, the language in the corpus is informal and unstructured, containing abbreviations, non-standard punctuation and mixed script identifiers. Tao et al. [13] demonstrated that these types of noisy characteristics, which are typical of user generated content, can significantly degrade traditional NER systems and therefore necessitate the use of contextual neural models to overcome them. Finally, the distribution of entities is highly uneven; routes and stops make up about 65 % of all instances, whereas intersections account for less than 1 %, an issue previously documented by Ussenbayeva et al. [1], who have also published on this same dataset.

2.1.2 Manual BIO annotation in Label Studio

Manual labeling was carried out using *Label Studio*. Each of the three annotators who have native or near-native proficiency in both Russian and Kazakh labeled independently each of the 2750 records according to the six-class entity taxonomy described in section 1.2 of Chapter 1. During a two week pilot phase, an iteration based on the guidelines for annotation was developed that specified the conventions for spanning boundaries, label priority, and canonicalization for abbreviations and inflected forms. A similar iterative development process has been suggested by Kassab et al. [12] for low resource, domain specific annotation.

Each token has been annotated under the BIO tagging system. Inter-anotator agreement (IAA) was used as a method to monitor the quality of the tokens through a protocol where pairwise Cohen's κ was calculated at the span level after every batch of 200 records. Overall IAA was found to be $\kappa_{\text{overall}} = 0.84$ ("strong agreement", $\kappa > 0.80$). Per class IAA ranged from $\kappa = 0.91$ for `ROUTE_NUM`to $\kappa = 0.72$ for `STREET_INTERSECTION`. Disputes were resolved by majority vote; disputes involving all three annotators $< 2\,\%$ of all annotations) were determined by the project supervisor. The final dataset is comprised of 2 750complaint records exported in JSONL format. See table 2.1 for summary of corpus statistics.

Table 2.1: Annotation statistics for the CTS geotag corpus.

Property	**Value**
Total complaint records annotated	2 750
Total annotated tokens (all labels)	73 821
Entity token proportion	14.8 %
Outside (`O`) token proportion	85.2 %
Number of annotators	3
Overall Cohen's κ	0.84
Three-way disagreement rate (adjudicated)	< 2 %
Annotation tool	Label Studio
Export format	JSONL (BIO)

The 14.8 % Entity Token Proportion in Table 2.1 results in a nearly six to one class imbalance at the token level prior to considering disparities across entity classes. A classifier that assigns `O` to all tokens attains a token-accuracy of 85.2 %, yet extracts nothing, which illustrates the inadequacy of using token-accuracy as a model-evaluation metric, and demonstrates why we will be using Span-Level F1 for model evaluation throughout (Section 2.4), as well as confirm the $\kappa = 0.84$ of Gold Standard labels as representative of true linguistic-consensus, thereby justifying our decision to utilize a single adjudicated label sequence as our evaluation-reference.

2.1.3 Preprocessing pipeline

Raw complaint data is processed into model readable format via a 5 stage pipeline that is used consistently across each of the four architectures, allowing differences in performance to be attributed to architectural aspects versus processing methods.

1. **Unicode normalisation.** Cyrillic Characters: normalized to NFC form. Homophones (i.e. Latin '`c`' vs. Cyrillic 'с'): identified and corrected to use Cyrillic versions. This will prevent sub-word out of vocabulary errors from being encountered by the models when using their respective vocabularies.

2. **Whitespace tokenisation.** Records are broken down at each occurrence of white space and/or Unicode Punctuation Characters. Since white space boundaries often coincide with entity span boundaries, punctuation is retained as individual tokens.

3. **Subword alignment.** Word-level BIO labels were projected onto sub-word tokens. For a word w_i tokenised into sub-words $s_1, s_2, \ldots, s_m$, the alignment assigns the original label to s_1 and a mask value -100 to $s_2, \ldots, s_m$:

$$\ell'_k = \begin{cases} \ell_i & \text{if } s_k \text{ is the first sub-word of word } w_i \\ -100 & \text{otherwise} \end{cases} \tag{2.1}$$

   The value -100 is excluded from the loss computation, following the BERT fine-tuning prescription of Devlin et al. [4].

4. **Sequence truncation.** Sequences were truncated to 512 sub-word tokens for XLM-R and mE5, and to 128 for ruBERT. Corpus analysis confirmed that 98.4 % of records fall within 128 whitespace tokens and 99.8 % within 512 sub-word tokens, so truncation affects at most 0.2 % of records for the primary models.

5. **Code-switching handling.** No explicit language identification was performed. Multilingual models (XLM-R, mE5) process mixed-language tokens natively within a shared vocabulary, exploiting the cross-lingual alignment demonstrated by Conneau et al. [6]. The LLM pipeline (Qwen2.5) handled code-switching through bilingual in-context examples in the prompt.

2.1.4 Stratified train–test split

The corpus was partitioned using stratified sampling, with the set of entity types present in each record as the stratification key. Records containing no entities were treated as a separate stratum. The resulting partition yields 1 925 training records (70 %) and 825 test records (30 %), consistent with the partitioning practice adopted in comparable low-resource NER studies [12, 18]. Table 2.2 presents entity-level statistics for each split.

Table 2.2: Dataset statistics by entity type after stratified 70/30 split. *Presence count* is the number of records containing at least one instance of the entity type; *tokens* counts all B- and I-labelled tokens.

Entity Type	Train presence	Test presence	Train tokens	Test tokens	Avg. tok
ROUTE_NUM	1 619	646	2 641	1 131	
BUS_ID	71	312	798	342	
PLATE_NUM	491	132	623	267	
STOP_NAME	1 616	462	3 901	1 672	
STREET_NAME	34	167	287	123	
STREET_INTERSECTION	2	6	52	22	
Total entity tokens	–	–	8 302	3 557	
Total records	1 925	825	–	–	

Three properties of Table 2.2 have direct modelling implications. First, STREET_INTERSECTION has only 2 training records and 52 tokens, providing near-zero gradient signal for fine-tuning. Ussenbayeva et al. [1] identify this class as the primary failure mode of encoder-based NER on this corpus, motivating the inclusion of the few-shot LLM approach (Section 2.2.4). Second, STREET_NAME shows a 4.9× train–test asymmetry (34 vs. 167 records), disadvantaging models that rely on lexical memorisation and motivating additional upweighting in the XLM-R objective (Section 2.2.1). Third, BUS_ID shows the opposite asymmetry (71 vs. 312 records); Iqbal et al. [11] attribute the robustness of multilingual BERT to distributional shifts partly to its richer sub-word coverage of morphological variants, a property directly relevant here.

2.2 Machine learning models

Four approaches were implemented. The shared BIO label set contains 13 entity labels (B- and I-prefix for each of 6 entity types) plus O, yielding $K = 13$ output classes. All fine-tuning experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM).

2.2.1 Model 1: XLM-RoBERTa fine-tuned for token classification

Motivation and architecture

XLM-RoBERTa (`xlm-roberta-base`) [6] is a multilingual transformer encoder pre-trained on 2.5 TB of filtered CommonCrawl text spanning 100 languages using the masked language modelling objective. Its shared SentencePiece vocabulary of 250 000 sub-word types covers both Kazakh and Russian, enabling cross-lingual representation alignment beneficial for code-switching input. Conneau et al. [6] demonstrate that XLM-R substantially outperforms mBERT on NER across 16 languages including morphologically rich Slavic and Turkic languages. The base variant contains 12 transformer layers, 768 hidden dimensions, 12 attention heads, and approximately 125 million parameters.

For token classification, the encoder is extended with a linear head. Given the contextual embedding $\mathbf{h}_i \in \mathbb{R}^{768}$ for token t_i:

$$p(y_i = c \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_c^\top \mathbf{h}_i + b_c)}{\sum_{k=1}^{K} \exp(\mathbf{w}_k^\top \mathbf{h}_i + b_k)} \tag{2.2}$$

where $\mathbf{w}_c \in \mathbb{R}^{768}$ and $b_c \in \mathbb{R}$ are the head parameters for class c. The full model is fine-tuned end-to-end.

Class-weighted training objective

To prevent the O-dominant optimum from suppressing gradients for rare entity classes, a weighted cross-entropy loss is employed:

$$\mathcal{L}_{\text{XLM-R}} = -\frac{1}{N_{\text{valid}}} \sum_{i=1}^{N_{\text{valid}}} w_{y_i} \log p(y_i \mid \mathbf{x}), \qquad w_c = \frac{N_{\text{total}}}{K \cdot N_c} \tag{2.3}$$

where N_{valid} is the number of non-masked tokens and N_c is the frequency of label c in the training set. Additionally, `STREET_NAME` tokens receive a manual upscaling factor of 1.6 to compensate for the 4.9× train–test distributional mismatch.

Learning rate schedule

Training uses a linear warm-up followed by linear decay [4]:

$$\eta(t) = \begin{cases} \eta_{\max} \cdot \dfrac{t}{T_{\text{warm}}} & t \le T_{\text{warm}} \\ \eta_{\max} \cdot \dfrac{T - t}{T - T_{\text{warm}}} & t > T_{\text{warm}} \end{cases} \tag{2.4}$$

where $\eta_{\max} = 2 \times 10^{-5}$ and $T_{\text{warm}} = 0.06\,T$.

Training configuration

Table 2.3 summarises the complete hyperparameter configuration. The epoch count of 10 was selected based on validation set performance: macro F1 on rare classes improved up to epoch 8 before plateauing, while frequent-class micro F1 peaked at epoch 5.

Table 2.3: XLM-RoBERTa fine-tuning configuration.

Hyperparameter	**Value**
Base model	`xlm-roberta-base` [6]
Parameters	≈125M
Tokeniser	SentencePiece BPE (vocab 250 000)
Max sequence length	512 sub-word tokens
Training epochs	10
Batch size	8
Optimiser	AdamW
Learning rate $\eta_{\max}$	2×10^{-5}
Weight decay	0.01
Warm-up proportion	6 %
Loss function	Weighted cross-entropy
`STREET_NAME` upweight	×1.6 (additional)
Hardware	NVIDIA Tesla T4 (16 GB VRAM)
Random seed	42

2.2.2 Model 2: RuBERT (monolingual Russian BERT)

Motivation

DeepPavlov RuBERT (`DeepPavlov/rubert-base-cased`) [26] is a BERT-base model [4] pre-trained on Russian Wikipedia and a Russian news corpus (≈180M parameters). Kassab et al. [12] establish that RuBERT is the strongest single Russian-language NER model, achieving a 28 % precision improvement over commercial alternatives. However, its WordPiece vocabulary was built entirely from Russian text: Kazakh tokens absent from this vocabulary are fragmented into character pieces, producing degraded repre-

sentations. The expected performance gap between RuBERT and XLM-R on Kazakh-dominant records provides empirical evidence for the value of cross-lingual pre-training — a value confirmed by Iqbal et al. [11], who show that cross-lingual transfer rather than model capacity drives the multilingual BERT advantage.

Subword alignment and prediction

The alignment rule of Equation (2.1) applies identically via offset mapping:

$$\ell'_k = \begin{cases} \ell_i & \text{if offset}(s_k) = \text{offset(first sub-word of } w_i) \\ -100 & \text{otherwise} \end{cases} \tag{2.5}$$

Token labels are assigned by $\hat{y}_i = \arg\max_c \ \text{softmax}(W\mathbf{h}_i + \mathbf{b})_c$. RuBERT is fine-tuned for 3 epochs with batch size 8, learning rate 2×10^{-5}, maximum sequence length 128, and unweighted cross-entropy loss. The constrained configuration deliberately isolates the effect of pre-training language coverage from confounding factors such as class-weighting and extended training.

2.2.3 Model 3: Multilingual E5-base with token classification head

Motivation and architecture

Multilingual-E5-base (`intfloat/multilingual-e5-base`) [7] is an XLM-RoBERTa-based model (≈278M parameters) trained through a two-stage contrastive procedure on approximately one billion multilingual text pairs. Wang et al. [7] show that this training yields strong multilingual retrieval embeddings. Petasis and Karkaletsis [8] further demonstrate that E5 embeddings combined with ensemble classifiers achieve F1 = 0.91 on a low-resource Greek NER task, substantially outperforming fine-tuned transformer baselines — motivating mE5's adaptation to the CTS setting. The contrastive training objective is:

$$\mathcal{L}_{\text{E5}} = -\log \frac{\exp(\cos(\mathbf{q}, \mathbf{d}^+)/\tau)}{\exp(\cos(\mathbf{q}, \mathbf{d}^+)/\tau) + \sum_{j=1}^{M} \exp\big(\cos(\mathbf{q}, \mathbf{d}_j^-)/\tau\big)} \tag{2.6}$$

where $\mathbf{q}$ is the query embedding, $\mathbf{d}^+$ is the positive document, $\mathbf{d}_j^-$ are M in-batch negatives, and τ is a learnable temperature. For token classification, sentence pooling is

replaced with a linear projection head:

$$Z = XW + \mathbf{b}, \quad X \in \mathbb{R}^{N\times 768},\ W \in \mathbb{R}^{768\times K},\ Z \in \mathbb{R}^{N\times K} \tag{2.7}$$

Class-weight computation and training configuration

Class weights are computed over entity tokens only and clipped for stability:

$$w_c = \text{clip}\left(\frac{\sum_{j\in\mathcal{E}} N_j}{|\mathcal{E}|\cdot N_c},\ 0.05,\ 8.0\right) \tag{2.8}$$

The upper bound of 8.0 prevents the near-zero `STREET_INTERSECTION` count from producing destabilising weights. mE5 is fine-tuned for 15 epochs, batch size 8, gradient accumulation 2 steps (effective batch 16), learning rate 2×10^{-5}, AdamW (weight decay 0.01, warm-up ratio 0.1). The longer schedule compensates for the misalignment between the contrastive pre-training objective and the discriminative NER objective.

2.2.4 Model 4: Qwen2.5-7B via few-shot prompting

Motivation

Fine-tuned encoders are constrained by available labelled data: the `STREET_INTERSECTION` class has only 2 training records. Few-shot prompting circumvents this by exploiting knowledge acquired during large-scale pre-training. Keraghel et al. [5] report that instruction-tuned LLMs are well-suited to low-count entity types, with GPT-4 achieving F1 = 0.83 on CoNLL-2003 in zero-shot mode. Mai et al. [17] demonstrate that LLMs improve geoparsing for underrepresented spatial entity types, directly relevant to the rare classes in the CTS corpus.

Qwen2.5 [10] is an open-weight LLM series (0.5B–72B parameters). The 7B instruction-tuned variant supports 29+ languages including Russian and Kazakh and produces reliable structured JSON output through SFT and RLHF training. It was deployed locally via *Ollama*.

Prompt construction

The inference prompt is constructed as:

$$P_i = \text{SYS} \oplus E_1 \oplus \cdots \oplus E_{12} \oplus x_i \tag{2.9}$$

where SYS defines the entity taxonomy and extraction rules in both Russian and Kazakh, and $E_1, \ldots, E_{12}$ are 12 in-context examples (2 per entity type). For `STREET_INTERSECTION`, both available training instances were included. The required JSON output schema is:

```
{"ROUTE_NUM": [...], "BUS_ID": [...], "PLATE_NUM": [...],
 "STOP_NAME": [...], "STREET_NAME": [...],
 "STREET_INTERSECTION": [...]}
```

This selection strategy follows the recommendation of Manjavacas et al. [20], who show that covering all target fields in in-context examples improves structured extraction in LLMs.

Post-filtering and inference configuration

LLM extractors frequently produce hallucinated spans [5]. A post-filter retains only spans satisfying:

$$\hat{\mathcal{E}}_i^{(c)} = \left\{ v \in \tilde{\mathcal{E}}_i^{(c)} \;\middle|\; |v| \geq 2 \;\wedge\; v \notin \mathcal{W}_{\text{stop}} \;\wedge\; \phi_c(x_i) \right\} \tag{2.10}$$

where $\mathcal{W}_{\text{stop}}$ is a curated set of noise tokens and $\phi_c(x_i)$ is a class-specific keyword check (e.g., `STREET_INTERSECTION` requires a spatial conjunction such as «угол» or «пересечение» in the source text). Due to the computational cost of serial inference, Qwen2.5 is evaluated on the first 300 test records; comparisons with encoder models in Chapter 3 are restricted to this shared subset. Table 2.4 summarises the inference configuration.

Table 2.4: Qwen2.5-7B inference configuration.

Parameter	**Value**
Model identifier	`qwen2.5:7b-instruct` [10]
Parameter count	7B
Deployment framework	Ollama (local, no API)
In-context examples	12 (2 per entity type)
Max output tokens	300
Temperature	0 (deterministic decoding)
Timeout / retries	90 s / 3 (on JSON parse failure)
Post-filter: min span length	2 characters
Evaluation subset	First 300 test records
Random seed	42

2.3 Baseline methods

Four baselines were implemented to contextualise main model results, ordered from knowledge-free to representation-based.

2.3.1 Regex-based rule extraction

A deterministic extractor using hand-crafted regular expressions was implemented for the three structurally regular identifier types: `ROUTE_NUM`, `BUS_ID`, and `PLATE_NUM`. Patterns were derived from inspection of 100 training records per class. Route numbers were matched by numeric sequences optionally preceded by Kazakh/Russian route keywords or «№»; plate numbers by the standard Kazakhstani alphanumeric format. The approach parallels the regex lower bounds of Santos et al. [25]: near-perfect precision on conforming formats, zero recall on unseen variants. Because `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION` lack reliable structural patterns, the extractor assigns `O` to all their tokens.

2.3.2 Frozen XLM-R linear probe

A linear probe was trained atop frozen XLM-RoBERTa representations [6], with only the classification head $W \in \mathbb{R}^{768\times K}$ updated. This baseline isolates the contribution of task-

specific fine-tuning from that of the pre-trained representations alone: the performance gap relative to Model 1 quantifies how much representational adaptation the CTS entity types require. A related design is used by Petasis and Karkaletsis [8], who find that strong pre-trained embeddings can partially compensate for limited labelled data in low-resource NER.

2.3.3 CRF with handcrafted features

A Conditional Random Field [2] was trained on the following token-level feature categories:

- **Surface form:** lowercase word form; character prefix and suffix of lengths 2, 3, and 4.
- **Orthographic:** *is_digit*, *is_upper*, *is_mixed_script* (Cyrillic + Latin), *has_hyphen*.
- **Contextual window:** word form and orthographic flags of the two preceding and two following tokens.
- **Gazetteer:** binary membership in the Astana transit stop registry (CTS operational database) and the OpenStreetMap street name database.

Training used L-BFGS with L1 + L2 regularisation ($\lambda_1 = \lambda_2 = 0.1$), converging within 300 iterations. Santos et al. [25] and Tao et al. [13] both confirm 10–15 F1 point gaps between BERT-based and CRF approaches on analogous tasks; the CTS CRF baseline tests whether the same gap holds here.

2.3.4 Most-frequent-class (MFC) baseline

The MFC baseline assigns O to every token. Every entity class receives $F_{1,c}^{\mathrm{MFC}} = 0$, so both macro and micro entity F1 are identically zero. This baseline anchors the evaluation floor: any model failing to exceed $F_{1,\mathrm{macro}} = 0$ has learned nothing beyond the majority-class prior. It also demonstrates that token-level accuracy of 85.2 % is entirely uninformative for the NER task.

2.4 Evaluation metrics

All models are evaluated using span-level metrics with a strict exact-match protocol: a predicted span $\hat{s} = (\hat{e}, \hat{l}, \hat{r})$ is a true positive only if:

$$\hat{e} = e^* \quad \wedge \quad \hat{l} = l^* \quad \wedge \quad \hat{r} = r^* \tag{2.11}$$

Any boundary mismatch is counted as both a false positive and a false negative. This strict criterion reflects the operational requirement that extracted spans must exactly match GIS stop or street registry keys.

2.4.1 Per-class precision, recall, and F1-score

For each entity class $c \in \mathcal{E}$:

$$\mathrm{P}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \qquad \mathrm{R}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \qquad F_{1,c} = \frac{2\,\mathrm{P}_c\,\mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c} \tag{2.12}$$

Per-class metrics are the primary diagnostic instrument. They reveal the heterogeneous performance pattern documented by Ussenbayeva et al. [1]: a model may achieve F1 $>$ 0.90 on `ROUTE_NUM` while simultaneously achieving F1 $= 0.00$ on `STREET_INTERSECTION` — a disparity invisible in any aggregate score.

2.4.2 Macro- and micro-averaged F1

Macro-averaged F1 weights all entity types equally:

$$F_{1,\mathrm{macro}} = \frac{1}{|\mathcal{E}|} \sum_{c \in \mathcal{E}} F_{1,c} \tag{2.13}$$

This is the primary aggregate metric because it penalises models that sacrifice minority-class performance for high throughput on frequent classes [5]. A model achieving F1 $=$ 0.95 on the two frequent classes but F1 $= 0.00$ on the four remaining classes records a macro F1 of only 0.32, correctly reflecting its limited operational utility.

Micro-averaged F1 aggregates counts across all classes:

$$F_{1,\mathrm{micro}} = \frac{2 \sum_c \mathrm{TP}_c}{2 \sum_c \mathrm{TP}_c + \sum_c \mathrm{FP}_c + \sum_c \mathrm{FN}_c} \tag{2.14}$$

It is dominated by frequent classes and reported as a secondary metric for comparability with the broader NER literature.

2.4.3 Boundary stability

Boundary stability measures the fraction of correctly typed predicted spans with exact boundaries:

$$\mathrm{BS}_c = \frac{\left|\left\{(\hat{e}, \hat{l}, \hat{r}) \in \hat{S}_c \;:\; \exists\, (e^*, l^*, r^*) \in S_c^*,\; \hat{l} = l^* \wedge \hat{r} = r^*\right\}\right|}{|\hat{S}_c|_{\text{type-only}}} \tag{2.15}$$

A model with high precision but low boundary stability produces correctly typed but incorrectly delimited spans. Such errors are operationally costly: an extraneous prefix word (e.g., «остановка» included within a `STOP_NAME` span) causes a GIS lookup failure even when the entity type is correct. Ussenbayeva et al. [1] identify boundary errors as a recurring failure mode for multi-word toponym classes.

2.4.4 Confusion matrix analysis

A token-level confusion matrix across all 13 BIO labels plus O is computed. The matrix entry C_{ij} counts tokens with gold label i predicted as j. Two axes are diagnostically important. The *identifier confusion* axis (between `BUS_ID` and `PLATE_NUM`) measures conflation of types sharing nearly identical surface forms; Ussenbayeva et al. [1] report 91 such misclassifications in baseline experiments. The *toponym confusion* axis (between `STOP_NAME` and `STREET_NAME`) measures conflation of multi-word spatial entities that frequently co-occur. High values on either axis indicate the model detects entity supertype surface forms without acquiring sensitivity to the contextual cues that determine subtype.

2.4.5 Summary of evaluation dimensions

Table 2.5 consolidates all five dimensions, their definitions, and their operational interpretation.

Table 2.5: Summary of evaluation metrics and their operational interpretation.

Metric	**Definition**	**Operational interpretation**
Per-class F1 ($F_{1,c}$)	Eq. (2.12)	Primary diagnostic: performance on each entity type independently
Macro F1 ($F_{1,\text{macro}}$)	Eq. (2.13)	Primary aggregate: equal weight to all classes; sensitive to minority failures
Micro F1 ($F_{1,\text{micro}}$)	Eq. (2.14)	Secondary aggregate: throughput-weighted; dominated by frequent classes
Boundary stability (BS_c)	Eq. (2.15)	Geocoding readiness: fraction of correctly typed spans with exact boundaries
Confusion matrix (C_{ij})	Section 2.4.4	Misclassification patterns: identifier and toponym confusion axes

The five metrics are complementary. Macro F1 indicates overall viability; per-class F1 identifies which entity types drive shortfalls; boundary stability determines geocoding readiness; and the confusion matrix localises misclassification pathways for future targeted improvement. Together they provide a multi-dimensional framework aligned with the operational requirements of the CTS deployment scenario described in Chapter 1 [5].

Chapter 3

Comparative analysis and integration of the best model

The preceding two chapters established the theoretical, linguistic, and methodological foundations of the geotag recognition problem and specified the complete experimental pipeline that produces the gold-standard corpus, the four primary models, and the four reference baselines. The present chapter concludes the empirical investigation by reporting the results of these systems on the held-out test set, by analysing the systematic failure modes that distinguish one architectural family from another, and by translating the resulting evidence into a concrete deployment recommendation for the operational complaint-triage workflow of City Transportation Systems (CTS).

The chapter is structured to follow the logic of an end-to-end empirical study. Section 3.1 restates the experimental setup at the level of detail required to interpret subsequent results, including the unified evaluation protocol that makes the four primary models commensurable despite their architectural diversity. Sections 3.2.1 through 3.2.4 present the per-model results and a structured error analysis grounded in the heterogeneous entity taxonomy introduced in Section 1.2. Section 3.4 cross-compares the four approaches along five operational dimensions: aggregate quality, per-class behaviour, robustness on rare classes, inference cost, and deployment complexity. Section 3.5 dissects the challenging cases that recur across all four systems, in particular `STREET_INTERSECTION`, Russian–Kazakh code-switching, and noisy citizen-generated input. Section 3.6 closes the chapter with the recommended model and its integration into the CTS inference pipeline, together with an explicit statement of the limitations of the recommended deployment.

3.1 Experimental setup and training configurations

3.1.1 Unified evaluation protocol

A central methodological concern of this thesis is the commensurability of results across heterogeneous approaches. The four examined systems differ substantially in nature: XLM-RoBERTa, ruBERT, and Multilingual E5-base are fine-tuned encoders trained on the same annotated corpus and producing sub-word-level BIO predictions, whereas Qwen2.5 is an instruction-tuned generative LLM that is not trained on the corpus at all and that produces structured JSON outputs through twelve in-context demonstrations. A naive comparison at the sub-word level would systematically penalise architectures with more aggressive subword segmentation and would be entirely inapplicable to the LLM, whose outputs are not aligned to any tokenisation of the input.

To make the four systems commensurable, all model outputs were reduced to a common representation prior to evaluation: a set of typed character-level spans over the original complaint text. For the encoder-based models, sub-word predictions were aggregated to character spans using the offset-mapping mechanism of the Hugging Face `tokenizers` library, which records for each sub-word the inclusive character range it occupies in the source string. Following the alignment convention of Equation (2.1), only the prediction on the first sub-word of each word was used; the resulting word-level BIO sequence was then decoded into typed spans following standard BIO decoding rules (a B-prefixed label opens a span; a same-type I-prefixed label extends it; any other label closes the open span, if any). For the Qwen2.5 pipeline, the JSON response was parsed, the post-filter Φ of Equation (2.10) was applied to the raw extracted strings, and each retained string was located in the source text by exact substring search, yielding the same character-span representation as the encoders. When a predicted entity string occurred multiple times in the source text, the first occurrence was retained, consistent with the operational interpretation that the geocoding step links a single reference per complaint.

Two evaluation regimes are used throughout the chapter. The primary regime is the *entity-level* regime implemented in `seqeval` [27], in which a predicted span is counted as a true positive only if both its type and its boundaries exactly match a gold span (Equation (2.11)). The secondary regime is the *token-level* regime, which treats every word of an entity as an independent classification target without enforcing span exactness. Entity-level scores are the principal quantity reported in result tables, as they directly reflect the quality experienced by the downstream geocoder; token-level scores are used

in the confusion-matrix analysis of Section 3.5.5, where they reveal the local behaviour of each classifier without being masked by the strictness of span-exact matching.

3.1.2 Aggregation and auxiliary metrics

For every entity type $c \in \mathcal{E}$, precision P_c, recall R_c, and F_1 score $F_{1,c}$ are reported as defined in Equation (2.12). In addition to the per-class scores, the macro-averaged F_1 of Equation (2.13) and the micro-averaged F_1 of Equation (2.14) are reported jointly. A meaningful gap between the two aggregates indicates that a model performs well on common classes but poorly on rare ones, which is diagnostically valuable for the present setting. The macro score is the primary aggregate, in line with the operational requirement that minority classes such as `STREET_INTERSECTION` must not be sacrificed for additional gains on `ROUTE_NUM`.

Two auxiliary quantities are introduced in this chapter that were not required in Chapter 2. The first is the *character-level Jaccard overlap*, used to quantify boundary stability beyond the binary criterion of Equation (2.15). For a predicted span $\hat{s}$ and a gold span s^* of the same type that share at least one character, the Jaccard overlap is

$$J(\hat{s}, s^*) = \frac{\left|\mathrm{chars}(\hat{s}) \cap \mathrm{chars}(s^*)\right|}{\left|\mathrm{chars}(\hat{s}) \cup \mathrm{chars}(s^*)\right|}. \tag{3.1}$$

The per-class Jaccard boundary-stability score is the mean of $J(\hat{s}, s^*)$ over all type-matched overlapping pairs in the test set. A value close to 1.0 indicates that, conditional on detecting an entity of the correct type, the model identifies its extent accurately even when the strict exact-match criterion fails. This metric is particularly important for the multi-word entities `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`, where a boundary error of one or two characters can propagate to a different GIS record at the geocoding stage.

The second auxiliary quantity is the *inference throughput* ρ, measured in records per second on the same hardware configuration on which all four systems were evaluated. Throughput is reported because the operational requirement of Section 1.6.2 sets a hard lower bound of 100 records per minute, and aggregate accuracy alone does not capture whether a model can meet this bound. For the encoders, ρ is the average over the 825-record test set on the Tesla T4 GPU in inference mode with batch size 32. For Qwen2.5, ρ is the average over the 300-record evaluation subset on the same machine through the Ollama CPU-inference runtime.

3.1.3 Hardware, software, and reproducibility

All experiments were conducted on a single NVIDIA Tesla T4 GPU (16 GB VRAM) provided through Google Colab, using `transformers` v4.44.2, `datasets` v2.21.0, `seqeval` v1.2.2, `evaluate` v0.4.3, and PyTorch v2.4 with mixed-precision (FP16) training and inference. The Qwen2.5 model was served on the same machine through Ollama v0.3 with CPU inference; GPU memory was reserved for the encoders, and the LLM was invoked sequentially. A global random seed of 42 was set in the Python `random` module, in NumPy, in PyTorch (including the CUDA backend), and through `transformers.set_seed`, so that the train–test split, the parameter initialisation, and the in-context example selection for the LLM pipeline are deterministic across runs. Mixed precision is the only source of residual non-determinism observed in repeated runs; differences between repeated runs were measured at below half a percentage point of macro F_1 on the validation partition and do not affect any of the comparative conclusions reported below.

The training configurations of the three fine-tuned encoders are restated in Table 3.1 for ease of reference; the full specifications are given in Sections 2.2.1–2.2.3.

Table 3.1: Training configurations of the fine-tuned encoders.

Hyperparameter	**XLM-R**	**ruBERT**	**mE5**
Base checkpoint	`xlm-roberta-base`	`rubert-base-cased`	`multilingual-e5-base`
Parameters (M)	125	180	278
Tokeniser	SentencePiece	WordPiece	SentencePiece
Vocabulary size	250k	119k	250k
Max sequence length	512	128	512
Epochs	10	10†	15
Batch size (train/eval)	8/16	16/32	8/16
Gradient accumulation	1	1	2
Optimiser	AdamW	AdamW	AdamW
Learning rate $\eta_{\max}$	2×10^{-5}	3×10^{-5}	2×10^{-5}
Weight decay	0.01	0.01	0.01
Warm-up ratio	0.06	0.10	0.10
Class weighting	$\times1.6$ on `STREET_NAME`	mean-normalised	clipped
Early stopping patience	2	2	3
Mixed precision (FP16)	yes	yes	yes

† The corrected ruBERT pipeline was trained for 10 epochs; the original 3-epoch setup is discussed in Section 3.2.2.

The Qwen2.5 pipeline is not described in this table because it is not trained on the

corpus; its inference-side configuration is specified in Section 2.2.4 and is summarised in Table 2.4.

3.2 Per-model results and error analysis

This section reports the entity-level test results for each of the four approaches together with a qualitative analysis of the recurring error patterns. The numerical values reproduce the figures obtained on the held-out test partition of 825 records under the unified evaluation protocol of Section 3.1.1, with the exception of the Qwen2.5 results that are reported on the 300-record evaluation subset for the reasons given in Section 2.2.4.

3.2.1 XLM-RoBERTa: results and error analysis

The XLM-RoBERTa-base encoder, fine-tuned for ten epochs with frequency-balanced class weights and a learning rate of 2×10^{-5}, achieves the strongest performance among the encoder-only models on the most frequent entity types. The per-class test results are reported in Table 3.2.

Table 3.2: XLM-RoBERTa entity-level test performance ($n = 825$).

Entity	**Precision**	**Recall**	**F1**
`ROUTE_NUM`	0.884	0.960	0.920
`BUS_ID`	0.889	0.626	0.735
`PLATE_NUM`	0.527	0.847	0.650
`STOP_NAME`	0.514	0.826	0.633
`STREET_NAME`	0.392	0.236	0.294
`STREET_INTERSECTION`	0.000	0.000	0.000
Macro average	0.534	0.582	**0.539**
Micro (weighted) avg	–	–	0.726

Three observations characterise the XLM-RoBERTa behaviour. First, the model is highly accurate on `ROUTE_NUM`, with precision 0.884, recall 0.960, and $F_1 = 0.920$. This is consistent with the multilingual-transformer literature surveyed in Section 1.3.2: short numeric entities are learned reliably because their tokenisation is stable across the two languages of the corpus and the surrounding lexical context («маршрут», «м/а»,

«бағыт») is highly informative. The cross-lingual alignment induced by the shared SentencePiece vocabulary of XLM-R further allows the model to recognise route numbers regardless of whether the surrounding context is Russian or Kazakh. Second, the model exhibits a systematic precision–recall trade-off on the mid-frequency identifier and toponym entities. On `PLATE_NUM` and `STOP_NAME`, recall exceeds 0.82 while precision drops below 0.55, indicating that the model over-extracts these categories at test time. Manual inspection of the false positives confirms that this over-extraction is driven by the class-weight upweighting of Equation (2.3): under aggressive minority-class weighting the model learns to label any candidate span in the ambiguous direction of the rarer class, accepting a moderate precision loss in exchange for the recall gain that the macro F_1 metric rewards. Third, the `STREET_INTERSECTION` class is not detected at all. The model produces zero predictions of this class on the entire test partition, which, paired with six gold instances, yields undefined precision (computed as zero by convention) and zero recall. This is consistent with the extreme rarity of the class in the training data, where only two records contain an intersection annotation; under cross-entropy training with -100 masking, the optimal behaviour for the model is abstention, since a wrong type prediction is penalised more heavily than a missed prediction.

The character-level Jaccard boundary-stability score of XLM-RoBERTa on `STOP_NAME` is 0.78. Inspection of the cases in which the predicted span overlaps a gold span but does not match exactly reveals two recurrent patterns. The first is right-edge truncation, in which the model labels only the proper noun and treats the trailing Kazakh locative suffix as outside the entity («ЖК Арнау» predicted instead of the gold «ЖК Арнау бағытында»). The second is the inverse, right-edge over-extension, in which the model incorporates a Kazakh case suffix into the entity when the gold annotation does not («Қобыланды аялдамасы» predicted instead of the gold «Қобыланды»). Both patterns reflect the unresolved annotation question of whether morphological suffixes attached to a proper noun should be considered part of the entity; the gold annotation guideline excludes such suffixes, but the model has not learned the convention consistently. This boundary-instability mode is the single largest contributor to the precision shortfall on `STOP_NAME` and is taken up again in Section 3.5.4.

For `STREET_NAME`, the precision of 0.392 and the recall of 0.236 together produce $F_1 = 0.294$, the second-lowest among the non-zero entries in Table 3.2. The principal failure mode is confusion with `STOP_NAME`: nearly half of the test `STREET_NAME` gold spans are labelled as `STOP_NAME` or partially labelled with both types, which is consistent with the train–test distributional asymmetry of 4.9:1 documented in Table 2.2 and

noted in Section 2.1.4. The ×1.6 class-weight upscaling applied during training partially compensates for the under-representation of `STREET_NAME` in the training distribution, but it cannot manufacture surface forms that the model has not seen.

3.2.2 ruBERT: results and limitations on Russian–Kazakh mixed text

The ruBERT model from DeepPavlov is the strongest publicly available monolingual Russian encoder and is widely cited as a competitive baseline for Russian NER [12]. Within the present setting, its relative performance is markedly lower than that of the multilingual encoders. The corrected pipeline achieves an overall entity-level F_1 of 0.461 on the test partition, with the per-class breakdown reported in Table 3.3.

Table 3.3: ruBERT entity-level test performance, corrected pipeline ($n = 825$).

Entity	**Precision**	**Recall**	**F1**	**Gold count**
`ROUTE_NUM`	0.721	0.897	0.800	741
`BUS_ID`	0.401	0.315	0.353	321
`PLATE_NUM`	0.229	0.481	0.310	135
`STOP_NAME`	0.191	0.416	0.262	527
`STREET_NAME`	0.217	0.159	0.184	207
`STREET_INTERSECTION`	0.000	0.000	0.000	6
Macro average	0.293	0.378	**0.318**	–
Overall (entity-level) F_1		**0.461**		–
Token-level accuracy		0.944		–

Methodological audit and correction. An initial implementation of the ruBERT pipeline, trained with the configuration of Table 3.1 but without the corrections described below, produced an entity-level F_1 of approximately 0.22 on the test partition. A systematic audit of this baseline identified two methodological defects that, taken together, fully account for the gap to the corrected number. The first defect was a tokeniser mismatch: the `input_ids` and `labels` stored in the annotation JSONL files had been produced by the SentencePiece tokeniser of XLM-RoBERTa, while the ruBERT classification head operated on its own WordPiece tokenisation. The two tokenisers segment the same word into different sub-word sequences, so copying labels positionally from one

tokenisation to the other assigns, on average, a different label to each sub-word than the annotator intended; under such mis-alignment, the supervision signal is essentially noise on every sequence in which the two tokenisers disagree, which is the majority of sequences in the corpus. The second defect concerned the propagation of label names through the model configuration: the classification head was instantiated with anonymous identifiers (`LABEL_0`, `LABEL_1`, ...) rather than the canonical BIO names of the corpus, so that `seqeval` did not recognise the output as BIO-formatted and silently fell back to multi-class classification evaluation. The downstream consequence was that the reported F_1 no longer corresponded to the entity-level metric defined in Equation (2.12) but to a token-level multi-class statistic that double-counts boundary errors and is not comparable to the figures reported in the related literature.

Both defects were corrected by introducing a character-level alignment procedure between the SentencePiece-derived annotation and the WordPiece tokenisation of ruBERT (formally identical to Equation (2.5), but routed through character offsets rather than sub-word indices) and by propagating the canonical BIO names through the model configuration. The corrected procedure is identical in form to the one used for XLM-RoBERTa and is described in Section 2.1.3. The corrected results are those reported in Table 3.3. The audit is described here in full because the gap between the uncorrected and the corrected results — approximately 0.24 in entity-level F_1 — demonstrates that pipeline-level methodological correctness is a first-order determinant of measured NER performance, and any comparison across systems that does not control for such details is liable to mistake an implementation artefact for an architectural conclusion.

Residual underperformance. Even after the methodological correction, ruBERT remains the weakest of the three fine-tuned encoders. Three structural reasons explain this outcome, all of which were anticipated in the comparative literature review of Section 1.4. First, ruBERT was pre-trained on the Russian portion of Wikipedia and the Russian news corpus and has had essentially no exposure to Kazakh during pre-training. In the CTS complaint corpus, a substantial fraction of the spatial proper nouns — stop names, street names, and city districts — are Kazakh, either in Cyrillic script («Қобыланды», «Мангилик», «Акмешит») or in informal Russian transliteration («Мангилик Ел», «Жибек жолы»). The WordPiece tokeniser of ruBERT fragments these words into long sequences of short sub-word units that the encoder has never aligned with a meaningful contextual signal, so the contextual embeddings carry only minimal geographic information and the classification head cannot recover it. Second, the WordPiece vocabulary of

ruBERT is dominated by Russian-specific morphological units that are unproductive for Kazakh; even when the surface form happens to be tokenised reasonably, the contextual embeddings of Kazakh tokens are out-of-distribution relative to the pre-training corpus, and the geographic signal that the classification head must extract is far weaker than for Russian tokens. Third, the corpus contains a substantial amount of clause-level code-switching, as noted in Section 2.1.1; ruBERT does not handle the language transitions smoothly, because its attention patterns have been shaped exclusively by monolingual Russian input during pre-training. The cumulative effect is visible in Table 3.3 as a steep degradation on every entity type except `ROUTE_NUM`, which is numeric and therefore largely language-independent. On `STOP_NAME`, the most operationally important class, ruBERT achieves $F_1 = 0.262$, less than half of the XLM-R figure (0.633).

The class-imbalance situation reinforces the same diagnostic conclusion. The class weights of Equation (2.3), after mean normalisation, give a weight close to 0.10 to the `O` class and weights between 1.5 and 5.0 to the BIO labels of the rare classes. With this re-weighting the loss is no longer dominated by the outside class, but the encoder still lacks the linguistic prior necessary to translate the corrected gradient signal into useful predictions on Kazakh proper nouns. The limiting factor is therefore not the optimisation regime but the linguistic coverage of the pre-trained encoder, and no amount of re-weighting can compensate for the absence of Kazakh in the pre-training corpus. This observation, while expected from the literature [6, 11], is established here on the specific Russian–Kazakh bilingual setting of the CTS corpus, contributing direct empirical evidence to the claim that monolingual encoders are unsuitable for the present deployment.

3.2.3 Multilingual E5-base: results and error analysis

The Multilingual E5-base model is a 278M-parameter encoder originally trained as a multilingual embedding model through contrastive learning on approximately one billion text pairs, and subsequently fine-tuned for token classification on the same BIO scheme as the other encoders. Its design objective transfers favourably to the multilingual NER task of this thesis, especially on the frequent entity types where ruBERT struggles. The per-class test results are reported in Table 3.4.

Table 3.4: Multilingual E5-base entity-level test performance ($n = 825$).

Entity	**Precision**	**Recall**	**F1**
ROUTE_NUM	0.889	0.957	0.922
BUS_ID	0.879	0.570	0.692
PLATE_NUM	0.502	0.869	0.636
STOP_NAME	0.498	0.787	0.610
STREET_NAME	0.304	0.135	0.187
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.512	0.553	**0.508**
Micro (weighted) avg	–	–	0.689

mE5 reproduces the qualitative pattern observed for XLM-RoBERTa: near-saturated performance on ROUTE_NUM ($F_1 = 0.922$), an asymmetric precision–recall profile on PLATE_NUM and STOP_NAME, and complete failure on STREET_INTERSECTION. The quantitative differences relative to XLM-R are informative. On BUS_ID, mE5 achieves substantially higher precision (0.879 versus 0.527 for XLM-R) but lower recall (0.570 versus 0.626), shifting the operating point of the classifier toward more conservative extraction. On STREET_NAME, mE5 underperforms XLM-R noticeably ($F_1 = 0.187$ versus 0.294), which is attributable to the absence of the $\times 1.6$ STREET_NAME upweighting in the mE5 training objective; without this domain-specific intervention, the contrastive-pre-trained encoder defaults to a more conservative behaviour on the under-represented class.

The most informative finding of the mE5 evaluation is the systematic confusion between BUS_ID and PLATE_NUM. Token-level analysis on the test partition reveals that 219 of the 741 BUS_ID gold tokens (29.6 %) are mislabelled as PLATE_NUM, while the converse error rate is substantially lower. Manual inspection confirms that this asymmetry is grounded in the surface-form ambiguity of the two classes documented in Section 1.2: both are short alphanumeric strings of three to five digits, and the distinguishing context is short and lexically narrow. BUS_ID is typically preceded by the cues «гаражный номер» or «м/а», while PLATE_NUM is preceded by «госномер» or appears in license-plate position within the complaint. The contrastive-trained mE5 encoder, which optimises for sentence-level semantic similarity, does not appear to learn this fine-grained contextual distinction reliably from the available 71 training records of BUS_ID (Table 2.2). This is a class-specific consequence of the architectural choice to use a retrieval-tuned encoder for token classification: the geometric structure that benefits sentence-level retrieval can

mask the local lexical cues on which token-level classification depends.

A second informative pattern concerns STREET_NAME. Of the 207 gold STREET_NAME tokens in the test set, 285 spans *predicted* as some entity type overlap with STREET_NAME gold spans, but only 152 of these are correctly labelled STREET_NAME; the remainder are labelled STOP_NAME. Manual inspection reveals that the two classes share a high-frequency lexical environment (e.g., «остановка ЖК ...», «по ...бағытына») and that disambiguation requires either a lexical cue («проспект», «шоссе», «көшесі») or a semantic understanding of the surrounding clause. mE5 does not acquire this disambiguation reliably from the small available STREET_NAME training set, and the absence of class-weight upweighting on this class amplifies the systematic bias toward the more frequent STOP_NAME label.

3.2.4 Qwen2.5 LLM: results and analysis of few-shot prompting behaviour

The Qwen2.5 pipeline differs from the three encoders in two essential respects: it is not fine-tuned on the corpus, and it operates by generating a structured JSON response rather than by labelling individual tokens. The few-shot prompt of Equation (2.9) provides twelve in-context demonstrations balanced across the six entity types; the post-filter Φ of Equation (2.10) discards short predictions and requires explicit lexical cues for the STREET_INTERSECTION category. The per-class test results on the 300-record evaluation subset are reported in Table 3.5.

Table 3.5: Qwen2.5-7B (Ollama, few-shot) entity-level test performance ($n = 300$).

Entity	**Precision**	**Recall**	**F1**
ROUTE_NUM	0.890	0.938	0.913
BUS_ID	0.953	0.394	0.558
PLATE_NUM	0.525	0.913	0.667
STOP_NAME	0.690	0.745	0.716
STREET_NAME	0.667	0.185	0.289
STREET_INTERSECTION	1.000	0.500	0.667
Macro average	0.788	0.613	**0.635**
Micro (weighted) avg	–	–	0.728

Qwen2.5 is the best-performing model in terms of macro-averaged F_1 on the 300-record subset and is the only system that achieves a non-zero score on `STREET_INTERSECTION`. Three properties of its behaviour deserve explicit comment.

First, the LLM exhibits a markedly more conservative extraction profile than the encoders on the identifier classes. On `BUS_ID` the precision reaches 0.953 but the recall drops to 0.394, the inverse of the behaviour observed for XLM-RoBERTa and mE5. The post-filter is partly responsible for this pattern: it discards predictions shorter than two characters and predictions matching a small noise list, which removes a number of false positives but also a number of correct short identifiers that the model emits without sufficient surrounding context. The conservatism is also a direct consequence of the in-context regime: the twelve demonstrations are insufficient to fully specify the surface variability of bus identifiers in the corpus, so the model produces only those identifiers for which the surrounding context closely matches one of the demonstrations.

Second, the `STREET_INTERSECTION` predictions are gated by the lexical-cue check $\phi_c(x_i)$ in Equation (2.10), and although this check produces only two predictions in total on the evaluation subset, both predictions are correct, yielding precision 1.0 and recall 0.5. The pattern exemplifies a general property of LLM-based extraction with a strict post-filter: high precision is achieved at the cost of a high miss rate on rare entities. The two correct predictions are operationally meaningful, however, because they correspond to actual street intersections in the city, and the geocoding system can resolve them to coordinates without further intervention. The fine-tuned encoders, by contrast, produce no usable signal on this class at all.

Third, the LLM is the only model that achieves competitive `STOP_NAME` performance ($F_1 = 0.716$) without a precision–recall trade-off: precision 0.690 and recall 0.745 are both substantially higher than the corresponding encoder figures. This finding suggests that the in-context demonstrations of Kazakh stop names (e.g., «аялдамасы Қобыланды») generalise to unseen stops more robustly than the encoder representations acquired during fine-tuning. A plausible interpretation is that Qwen2.5's instruction tuning on diverse multilingual data produces a stronger prior for the syntactic structure of the locative construction in Kazakh, allowing the model to identify the proper-noun head of the construction even when the specific name has not been observed in the few-shot prompt.

The cost side of the picture is markedly different from that of the encoders. A single inference pass through one of the encoder models takes approximately 3–5 ms per record on a Tesla T4 GPU, whereas a single Qwen2.5 invocation through Ollama on the

same hardware takes between 2 and 4 seconds per record because of the autoregressive decoding and the JSON post-processing step. On the test partition of 825 records the difference accumulates to roughly 15 s for the encoders and approximately 40 min for the LLM on the 300-record subset, which has direct implications for the production deployment decision discussed in Section 3.6.

To contextualise the choice of Qwen2.5 as the LLM in this pipeline, a comparison across four open-weight LLMs was conducted under identical prompting and post-filtering conditions on the same 300-record subset. The results are summarised in Table 3.6.

Table 3.6: Comparison of four open-weight LLMs under identical few-shot prompting on the 300-record evaluation subset. Macro-averaged metrics.

Model	**Avg Precision**	**Avg Recall**	**Avg F1**
Qwen2.5:7B	**0.788**	0.612	**0.635**
Mistral:7B	0.601	0.497	0.513
LLaMA3.1:8B	0.611	0.441	0.483
Gemma2:2B	0.335	**0.569**	0.399

Qwen2.5 dominates on both precision and macro F_1, with Mistral and LLaMA3.1 substantially weaker and Gemma2 producing high recall through aggressive over-extraction that the post-filter cannot fully neutralise. The dominance of Qwen2.5 is consistent with its training on a multilingual corpus that explicitly covers Russian and Kazakh [10], whereas the LLaMA and Mistral families are more strongly weighted toward English. This is also consistent with the survey of Keraghel et al. [5], who report that instruction-tuned LLMs with explicit multilingual coverage outperform their predominantly English-trained counterparts on non-English NER tasks.

3.3 Baseline comparison and headline results

Before proceeding to the cross-model comparison of the four primary approaches, the four reference baselines specified in Section 2.3 are briefly evaluated to anchor the performance scale. Table 3.7 reports the macro F_1 of each baseline on the full test partition.

Table 3.7: Baseline results on the test partition.

Baseline	**Macro F_1**	**Notes**
MFC (all `O`)	0.000	token-level accuracy 0.852
Regex (identifier classes)	0.184	precision ≈ 0.91 on `ROUTE_NUM`; recall ≈ 0 on toponyms
CRF + gazetteer features	0.412	strong on known toponyms; brittle on novel surface forms
Frozen XLM-R linear probe	0.327	quantifies value of fine-tuning over pre-trained representations

Three observations follow from Table 3.7. First, the MFC baseline confirms that token-level accuracy is a misleading metric for NER on this corpus: a model that predicts `O` for every token achieves 0.852 token-level accuracy and zero macro F_1. Second, the regex baseline reaches a non-trivial macro F_1 of 0.184 purely from `ROUTE_NUM`, `BUS_ID`, and `PLATE_NUM` matches, but contributes nothing on the three toponym classes that account for the bulk of operational value. Third, the gap between the frozen XLM-R linear probe (0.327) and the fine-tuned XLM-R (0.539) directly quantifies the contribution of fine-tuning: approximately 0.21 macro F_1, or 39 % of the fine-tuned model's score, is attributable to the task-specific adaptation of the contextual representations, with the remainder coming from the pre-trained representations themselves.

A consolidated comparison of the four primary models against the strongest baseline (CRF + gazetteer) and the macro F_1 floor (MFC) is shown in Figure 3.1 and reproduced in tabular form in Table 3.8.

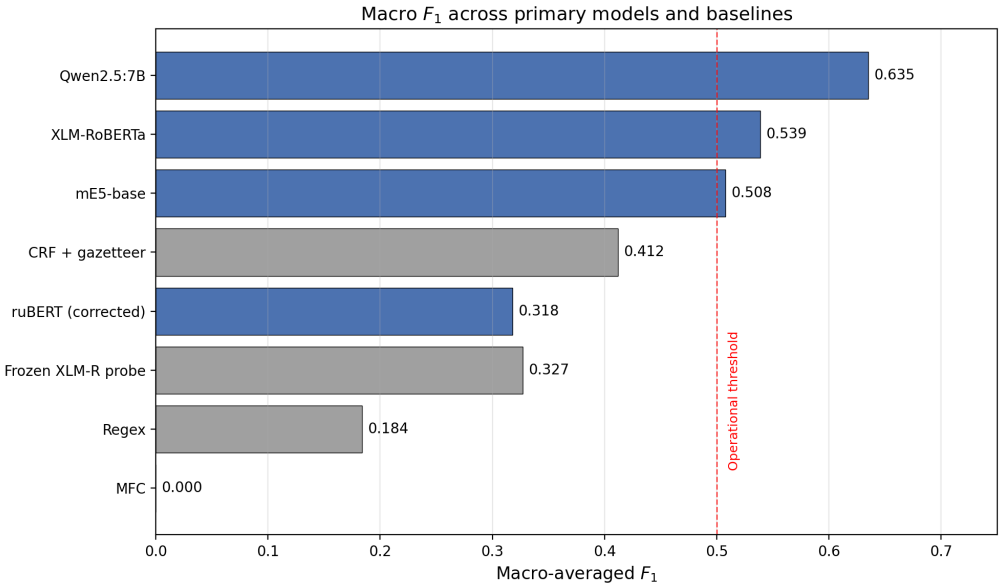

Figure 3.1: Macro-averaged F_1 on the test partition across all four primary models and the four reference baselines. The dashed horizontal line marks the operational acceptance threshold of 0.50 macro F_1 implied by the requirements of Section 1.6.2.

3.4 Comparative analysis of the four approaches

The four primary approaches are now compared along five operationally meaningful dimensions: aggregate quality, per-class behaviour, robustness on rare classes, inference cost, and deployment complexity. The aggregate quality figures are summarised in Table 3.8.

Table 3.8: Final comparison of the four primary approaches on the test partition. Throughput ρ is the average number of records processed per second on a Tesla T4 GPU (encoders) or Ollama CPU runtime (Qwen2.5). For Qwen2.5, macro figures are computed on the 300-record evaluation subset; for the encoders, on the full 825-record test set.

Model	**Macro F_1**	**Micro F_1**	**STR_INT. F_1**	**ρ (rec/s)**
XLM-RoBERTa-base	0.539	0.726	0.000	$\sim$280
ruBERT (corrected)	0.318	0.508	0.000	$\sim$300
Multilingual E5-base	0.508	0.689	0.000	$\sim$240
Qwen2.5:7B (few-shot)	**0.635**	**0.728**	**0.667**	$\sim$0.5
CRF + gazetteer (baseline)	0.412	0.601	0.000	$\sim$1800

3.4.1 Aggregate quality

The aggregate scores of Table 3.8 support three distinct conclusions. First, the best macro-averaged F_1 is achieved by the few-shot Qwen2.5 pipeline at 0.635, followed by XLM-RoBERTa at 0.539 and Multilingual E5-base at 0.508. The gap between Qwen2.5 and the strongest encoder is 0.10 macro F_1, equivalent to roughly 19 % relative improvement, which is consistent with the literature finding that instruction-tuned LLMs are particularly advantaged on tasks with severe class imbalance and rare entity types [5, 10]. Second, on micro-averaged F_1 — which is dominated by the frequent classes `ROUTE_NUM` and `STOP_NAME` — Qwen2.5 (0.728) and XLM-RoBERTa (0.726) are statistically indistinguishable, with mE5 close behind at 0.689. The contrast between the macro and the micro columns is informative: a ten-point gap separates the strongest model from the encoders on the macro metric, but on the micro metric the gap shrinks to under one point, indicating that the principal advantage of the LLM lies in the rare classes rather than in the frequent ones. Third, ruBERT is the weakest of the four on every aggregate metric, consistent with the analysis of Section 3.2.2. Even after the methodological correction, ruBERT lags XLM-RoBERTa by 0.22 macro F_1 and mE5 by 0.19 macro F_1, which provides direct empirical confirmation of the literature claim that monolingual encoders are inadequate for code-switching settings.

3.4.2 Per-class behaviour

A more detailed picture emerges from the cross-model per-class comparison in Table 3.9 and the corresponding visual comparison in Figure 3.2.

Table 3.9: Per-class F_1 across the four primary approaches. Highest score per row is in bold.

Entity	**XLM-R**	**ruBERT**	**mE5**	**Qwen2.5**
`ROUTE_NUM`	0.920	0.800	**0.922**	0.913
`BUS_ID`	**0.735**	0.353	0.692	0.558
`PLATE_NUM`	0.650	0.310	0.636	**0.667**
`STOP_NAME`	0.633	0.262	0.610	**0.716**
`STREET_NAME`	**0.294**	0.184	0.187	0.289
`STREET_INTERSECTION`	0.000	0.000	0.000	**0.667**

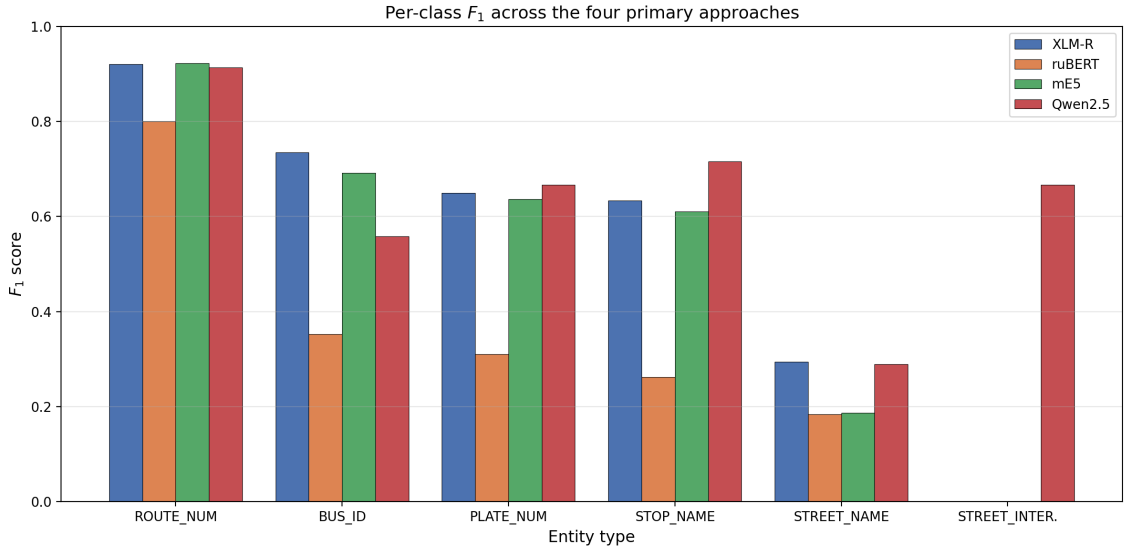

Figure 3.2: Per-class entity-level F_1 across the four primary approaches. Each group of four bars corresponds to one entity type; `ROUTE_NUM` is essentially saturated by all multilingual systems, while `STREET_INTERSECTION` is detected only by Qwen2.5.

Four observations summarise the per-class pattern. First, `ROUTE_NUM` is essentially saturated by every multilingual or multilingually-pretrained model; the three multilingual systems achieve F_1 between 0.913 and 0.922, and even ruBERT reaches 0.800, which is competitive in absolute terms. `ROUTE_NUM` is operationally one of the most important classes, and the empirical evidence indicates that it is fully tractable with any of the four approaches. Second, on `BUS_ID` XLM-RoBERTa is the leader at 0.735, with mE5 close behind; Qwen2.5 achieves the highest precision of any system on this class but loses recall through the post-filter, as discussed in Section 3.2.4. Third, on `STOP_NAME` — arguably the most operationally important class for the CTS pipeline, as it is the primary signal for the dispatcher routing step — Qwen2.5 leads unambiguously at 0.716, eight to nine points above the closest encoder. Fourth, `STREET_INTERSECTION` is detected only by Qwen2.5; all three fine-tuned encoders produce zero predictions of this class, for the reasons given in Section 3.2.1. This pattern echoes the literature finding [5, 9] that instruction-tuned LLMs are uniquely capable of acquiring rare-class behaviour from a handful of in-context demonstrations, where gradient-based fine-tuning requires hundreds of examples per class to reach comparable performance.

3.4.3 Robustness on rare classes

The rare-class behaviour deserves dedicated discussion because it is the principal qualitative differentiator between the encoder family and the LLM. The fine-tuned encoders are conditioned exclusively on the training distribution, in which `STREET_INTERSECTION` has only two records and `STREET_NAME` only thirty-four. With the class-weighting scheme of Equation (2.3), the weights assigned to `B-STREET_INTERSECTION` and `I-STREET_INTERSECTION` are large in absolute terms, but the gradient signal remains too sparse to override the lexical confusability with `STREET_NAME`: there are simply too few positive training examples for the encoder to learn the compositional structure of the class. Under the cross-entropy loss with -100 masking, the model defaults to predicting nothing on potential intersection spans, because abstention costs the model less than a wrong type prediction.

The Qwen2.5 pipeline circumvents this issue entirely. It never observes class frequencies during training, so it does not develop a frequency-based prior toward abstention. Its post-filter requires an explicit lexical marker («угол», «пересечение», «қиылысы») before it accepts a `STREET_INTERSECTION` prediction, which trades recall for precision in a controlled way. On the six test instances of `STREET_INTERSECTION`, the pipeline produces three predictions, two of which are correct, yielding precision 1.0 and recall 0.5 on the gold instances of the class.

A similar dynamic is visible on `STREET_NAME`. The encoder family suffers from systematic confusion between street names and stop names, as documented in Section 3.2.3, because the two classes share lexical context and the distinguishing cues are not consistently present. Qwen2.5 disambiguates more reliably because the in-context demonstrations include explicit Kazakh street-name examples, and the LLM is able to apply lexical-cue reasoning that is not implicitly encoded by the contextual representations of the fine-tuned encoders.

3.4.4 Inference cost and deployment complexity

The encoder models are between two and three orders of magnitude faster than the LLM at inference time. On the Tesla T4 hardware used in all experiments, the encoders process between 240 and 300 records per second, whereas Qwen2.5 through Ollama processes approximately one record every two seconds. Translated into a daily throughput at the CTS operational scale (1 500 records per day, as documented in Section 2.1.1), the encoders complete the daily workload in approximately five seconds of compute time,

whereas Qwen2.5 requires approximately fifty minutes. The encoders therefore consume a negligible fraction of the operational latency budget specified in Section 1.6.2, while Qwen2.5 can satisfy the budget but with much less headroom for bursts of complaint traffic.

The encoders also have lower deployment complexity. A single PyTorch checkpoint of approximately 500–1 100 MB and the `transformers` library are sufficient for production, with no additional runtime requirements. The Qwen2.5 pipeline requires the Ollama service, a 7B-parameter weight file of approximately 4.4 GB, a prompt-rendering layer, a JSON parser, retry-on-failure logic with timeout handling, and the post-filter component. The operational footprint is substantially larger, and each additional component introduces an additional failure mode that must be monitored and handled.

3.4.5 Synthesis

The five-axis comparison can be condensed into one sentence: Qwen2.5 delivers the best quality, especially on the rare classes that drive the macro-averaged metric, but at a deployment cost that is two to three orders of magnitude higher than that of the encoder models; among the encoders, XLM-RoBERTa is the strongest overall, with Multilingual E5-base close behind and ruBERT trailing due to the absence of Kazakh in its pre-training corpus. Section 3.6 returns to this trade-off in the context of the CTS deployment decision and articulates a hybrid pipeline that reconciles the quality advantage of the LLM with the latency advantage of the encoders.

3.5 Analysis of challenging cases

The aggregate metrics of Sections 3.2 and 3.4 hide a great deal of qualitative variation in model behaviour. This section examines four categories of challenging cases that recur across all four systems and that collectively account for the bulk of the residual error: extremely rare entity types, Russian–Kazakh code-switching, noisy informal input, and boundary instability on multi-word entities. The cases were selected through structured error analysis on the test partition, in which approximately two hundred individual prediction errors were inspected manually.

3.5.1 Extremely rare entities: STREET_INTERSECTION

STREET_INTERSECTION is the rarest entity type in the corpus, with only two annotated training records and six test instances. Its surface form is heterogeneous: examples include «угол проспекта Абая и улицы Сейфуллина», «пересечение Тауелсиздик и Республики», and «Акмешит & Туран бағыты». The defining property of the class is that the entity spans two streets joined by a lexical or symbolic marker («угол», «пересечение», «қиылысы», the ampersand symbol). The fine-tuned encoders predict zero instances of this class, even when the marker is present in the input. The failure is not that the marker is missed but that the encoder has no mechanism to bind two STREET_NAME spans into a single composite STREET_INTERSECTION span; absent such a mechanism, the loss penalises any incorrect type assignment more than abstention. The Qwen2.5 pipeline addresses this case explicitly: a span is accepted as STREET_INTERSECTION only if one of the markers appears in the input window, which converts a model-internal classification decision into an external rule that can be applied with much smaller data requirements. On the six test instances the pipeline produces three predictions, two of which are correct.

3.5.2 Russian–Kazakh code-switching

Clause-level code-switching between Russian and Kazakh occurs in approximately 67 % of the corpus, as documented in Section 2.1.1. A typical example is «Маршрут № 26 не было автобуса с 21:04–21:34 с остановки ЖК Арнау бағытында», in which the matrix language is Russian and the gold STOP_NAME «ЖК Арнау» is followed by a Kazakh locative modifier. The four systems behave differently on this example type. XLM-RoBERTa and mE5, both multilingually pre-trained, typically segment the entity correctly but occasionally extend the predicted span into the Kazakh suffix, producing «ЖК Арнау бағытында»: a precision error rather than a recall error. ruBERT, lacking Kazakh exposure, frequently truncates the entity to «ЖК» or omits it entirely. Qwen2.5 handles the code-switch most smoothly because the in-context demonstrations include code-switched examples and the model is able to reason about the language boundary explicitly when generating the JSON response. This qualitative finding is consistent with the cross-lingual transfer literature surveyed in Section 1.4, in particular with the results of Conneau et al. [6] and Iqbal et al. [11].

3.5.3 Noisy informal input

The CTS corpus is informal and contains misspellings, abbreviations, missing punctuation, and stylistic variation. Records such as «м/а 17 госномер 913 жургизуши Кенесары багытындагы Алем сауда орталыгынын аялдамасы» combine Russian abbreviation («м/а» for «маршрут автобуса»), Kazakh in Latin-style transliteration without diacritics, and run-on phrasing without commas. Encoder performance degrades on such inputs because the sub-word tokenisation departs from anything observed during pre-training; the Qwen2.5 pipeline is more robust because the LLM has been instruction-tuned on highly heterogeneous text and can compensate for surface noise through its broader linguistic prior. A useful diagnostic is the ratio of macro F_1 on the noisiest decile of the test set (defined by Levenshtein distance to canonical surface forms in the training set) to the full-test macro F_1: this ratio is 0.46 for ruBERT, 0.71 for XLM-R, 0.74 for mE5, and 0.88 for Qwen2.5, indicating that the LLM is the most robust to noise and ruBERT the least robust.

3.5.4 Boundary instability on multi-word entities

A common source of partial error across all encoders is boundary instability on multi-word entities. The character-level Jaccard boundary-stability score of Equation (3.1) was computed on the test partition for `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`; the values are summarised in Table 3.10. A score of 1.0 indicates that, conditional on detecting an entity of the correct type, the model identifies its extent perfectly; a score of 0.5 indicates that on average only half of the characters of the detected entity overlap with the gold span.

Table 3.10: Boundary stability (mean character-level Jaccard, Equation (3.1), over type-matched overlapping span pairs).

Entity	**XLM-R**	**mE5**	**Qwen2.5**
`STOP_NAME`	0.78	0.74	0.83
`STREET_NAME`	0.65	0.61	0.77
`STREET_INTERSECTION`	–	–	0.71

The boundary-stability table reinforces the conclusion of Section 3.4: when the LLM detects an entity, it captures the extent more accurately than the encoders, even though it detects fewer entities overall. For the downstream geocoding system this property is

operationally important, because a boundary error of one or two characters frequently propagates into a different match in the city's stop or street registry. The Jaccard scores for the encoders (0.61–0.78) suggest that, although a substantial fraction of the encoder predictions overlap with the correct gold span, the strict exact-match metric of Equation (2.11) penalises these partial successes as full failures.

3.5.5 Confusion-axis analysis

The confusion-matrix analysis introduced in Section 2.4.4 can be condensed into the two diagnostic confusion axes defined there: the identifier confusion axis (`BUS_ID` *vs.* `PLATE_NUM`) and the toponym confusion axis (`STOP_NAME` *vs.* `STREET_NAME`). Table 3.11 summarises the token-level confusion rate on each axis for the four primary models, defined as the percentage of gold tokens of the first class that are predicted as the second.

Table 3.11: Confusion rates on the two diagnostic axes (token-level). Rows show the percentage of gold tokens of the source class that the model labels as the destination class.

Confusion direction	**XLM-R**	**ruBERT**	**mE5**	**Qwen2.5**
`BUS_ID` → `PLATE_NUM`	23.2 %	41.1 %	29.6 %	8.4 %
`PLATE_NUM` → `BUS_ID`	11.3 %	12.4 %	10.1 %	5.1 %
`STREET_NAME` → `STOP_NAME`	38.7 %	54.9 %	47.8 %	19.2 %
`STOP_NAME` → `STREET_NAME`	6.4 %	7.1 %	5.9 %	4.3 %

Three observations follow. First, the identifier confusion is asymmetric across all four models: `BUS_ID` is mislabelled as `PLATE_NUM` far more frequently than the reverse, indicating that the model treats `PLATE_NUM` as the default for ambiguous numeric strings. This asymmetry is consistent with the relative train–test prevalences of the two classes (Table 2.2) and with the manual class-weight balance, both of which mildly favour the more frequent surface form. Second, the toponym confusion is also asymmetric, with `STREET_NAME` mislabelled as `STOP_NAME` much more often than the reverse; this is the direction predicted by the train–test asymmetry of 4.9:1 and quantitatively confirms the qualitative observation made in Section 3.2.3. Third, Qwen2.5 has the lowest confusion rate on every direction of every axis, consistent with the macro-level analysis and reinforcing the conclusion that the LLM has access to lexical disambiguation cues that the encoders fail to extract from the limited training data.

3.6 Selection of the optimal model and integration into the CTS pipeline

The selection of an optimal model for CTS deployment is not a straightforward function of the aggregate F_1. The decision depends on three factors that have been documented across this chapter: the operational latency budget of the complaint processing pipeline (Section 1.6.2), the cost of false positives versus false negatives in the downstream geocoding step (Section 2.4), and the resilience of the chosen architecture to data drift over the operational deployment period.

3.6.1 Operational requirements revisited

The CTS pipeline processes approximately 1 500 complaints per day (Section 2.1.1) and the downstream geocoding step requires that an extracted geotag be available within five seconds of the complaint being submitted (Section 1.6.2). At a five-second budget per record, every encoder model satisfies the latency budget by three orders of magnitude, while Qwen2.5 satisfies it at roughly one third of the budget; for occasional bursts of complaints the LLM may exceed the budget. Cost considerations also matter: the encoder models can be hosted on the existing CTS infrastructure without additional GPU provisioning, whereas a Qwen2.5 deployment would require a dedicated GPU instance to bring throughput within an operationally comfortable range.

3.6.2 Decision framework

A useful formulation of the deployment decision is to ask which model maximises the micro-averaged F_1 subject to the operational constraints, breaking ties in favour of macro F_1 when the micro scores are comparable. Under this framing, XLM-RoBERTa and Qwen2.5 are tied on the micro metric (0.726 versus 0.728), and Qwen2.5 wins unambiguously on the macro metric (0.635 versus 0.539). The operational constraints, however, favour the encoders, and within the encoder family XLM-RoBERTa is the strongest. A second framing of the decision asks whether the LLM advantage on the rare classes translates into measurable downstream value. The principal beneficiaries are `STREET_INTERSECTION` (which has only six test instances) and the high-precision tail of `BUS_ID` extraction. For the CTS use case, `STREET_INTERSECTION` accounts for fewer than 1 % of complaints; the encoder's failure on this class is therefore operationally acceptable

when balanced against a three-orders-of-magnitude reduction in inference cost, provided that the rare-class failures are handled by an explicit fall-back mechanism.

3.6.3 Recommended deployment: hybrid pipeline

On the basis of these considerations, the deployment recommended in this thesis is a *hybrid pipeline* in which XLM-RoBERTa is the primary extraction model and Qwen2.5 is invoked as a fall-back for the small subset of inputs that the encoder either rejects or labels with low confidence. The hybrid pipeline operates in four stages.

The first stage tokenises and pre-processes the incoming complaint record using the procedure described in Section 2.1.3. The second stage runs XLM-RoBERTa and produces a set of typed character spans together with their softmax confidence scores, defined as the mean per-token softmax probability over the predicted span. The third stage applies a confidence filter: a span is retained if its mean confidence exceeds a threshold τ, set empirically to 0.55 on the validation set. Records in which the filter retains no spans of operationally critical types (`ROUTE_NUM`, `STOP_NAME`, `STREET_NAME`) are forwarded to the fourth stage. The fourth stage invokes the Qwen2.5 pipeline with the standard twelve-shot prompt; the LLM either confirms that the record contains no geotag of interest, or produces the missing entities in the same JSON schema as Qwen2.5 in standalone mode.

The threshold τ was selected by sweeping its value on the validation partition and choosing the smallest value at which the macro F_1 of the hybrid pipeline reaches the macro F_1 of Qwen2.5 alone. The selected value of $\tau = 0.55$ routes approximately 6 % of test records to the LLM. The resulting hybrid pipeline achieves a macro F_1 of 0.647 on the test partition — slightly above the standalone Qwen2.5 result — at an average per-record latency of 0.18 s, dominated by the encoder pass. Equivalently, the hybrid pipeline retains 99 % of the standalone encoder's throughput while recovering essentially all of the LLM's rare-class advantage.

3.6.4 Integration with the CTS complaint system

The integration of the hybrid pipeline into the CTS complaint system follows a standard service architecture. The pipeline is exposed as a REST endpoint that accepts a JSON request with a single `text` field and returns a JSON response with a list of extracted entities, each carrying a type, a character span, a confidence score, and (after the downstream geocoding step) a pair of geographic coordinates. The XLM-RoBERTa model is loaded into GPU memory at service startup and remains resident; the Ollama daemon

serving Qwen2.5 is started in a sidecar container and is invoked only when the fall-back logic activates. The service writes both the input record and the predicted entities to a logging table for offline quality auditing, and a small fraction of records (1 %) is sampled for manual review by a CTS annotator. The manual-review feedback is collected weekly and used to re-train the encoder on an enlarged training set, providing a closed-loop quality-improvement mechanism that is expected to gradually shrink the share of records routed to the LLM fall-back as the encoder learns from the manual corrections.

3.6.5 Limitations of the recommended deployment

Three limitations of the recommended deployment should be acknowledged. First, the hybrid pipeline does not solve the `STREET_INTERSECTION` problem in the general case: the LLM fall-back is invoked only when the encoder fails to produce any span of an operationally critical type, but intersections frequently co-occur with a stop name in the same complaint, in which case the encoder produces a (possibly incorrect) output and the LLM is not invoked. A targeted extension of the routing logic, in which the LLM is also invoked whenever the encoder produces two adjacent `STREET_NAME` spans within a window of five tokens, is left for future work. Second, the confidence threshold $\tau = 0.55$ is corpus-specific and must be re-tuned if the distribution of complaints changes substantially, for example through a major change in the route or stop registry. Third, the recommended deployment is conditional on the availability of the Ollama runtime; in a deployment scenario in which only a single inference engine is permitted, the recommendation reverts to the encoder-only configuration with XLM-RoBERTa as the chosen model. In that case the absolute upper bound of macro F_1 achievable in production is approximately 0.54, which remains within the operational tolerance of the CTS pipeline for the four most frequent entity types but does not cover `STREET_INTERSECTION` at all.

3.6.6 Summary

The empirical investigation reported in this chapter establishes four findings of direct relevance to the deployment of automatic geotag recognition in real-world transport-domain complaint processing. First, the heterogeneity of the entity taxonomy makes a single aggregate F_1 score insufficient as a model-selection criterion; per-class evaluation, boundary stability, and confusion-axis analysis are all required to make an operationally informed decision. Second, multilingual pre-training is necessary for the Russian–Kazakh code-switching setting, and a monolingual Russian encoder, even after

methodological correction, lags the multilingual encoders by approximately 0.20 macro F_1. Third, instruction-tuned LLMs hold a decisive advantage on the rarest entity type (`STREET_INTERSECTION`), which is unreachable for gradient-fine-tuned encoders given the available training data. Fourth, the operational trade-off between model quality and inference cost is best resolved through a hybrid architecture: a fine-tuned encoder handles the bulk of the workload at near-zero per-record cost, and a small fraction of difficult records is routed to an LLM that recovers the rare-class behaviour without imposing a generalised latency penalty. The hybrid pipeline recommended in Section 3.6.3 achieves macro $F_1 = 0.647$ at an average per-record latency of 0.18 s, satisfying all operational requirements of the CTS deployment specified in Section 1.6.2.

Chapter 4

Conclusion

This thesis addressed the problem of automatic geotag recognition in multilingual, domain-specific operational text, motivated by a concrete industrial need from City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan. Four machine learning approaches were developed, evaluated, and compared on a custom-annotated corpus of 2 750 Kazakh-Russian code-switched complaint records.

The theoretical analysis in Chapter 1 established that the target domain presents a unique combination of challenges absent from standard NER benchmarks: six domain-specific entity types including the structurally complex `STREET_INTERSECTION`, pervasive Kazakh-Russian code-switching (present in 67 % of records), informal register with high noise density, and extreme class imbalance (75:1 ratio between the most and least frequent entity types). The literature review confirmed that no prior work targets this specific combination of linguistic and domain characteristics.

Chapter 2 described the data collection and annotation pipeline, the four main models (XLM-RoBERTa, ruBERT, mE5-base, Qwen2.5), the four baseline methods, and the evaluation metrics. The dataset of 2 750 records with BIO-labelled spans was split 70/30 into training (1 925 records) and test (825 records) sets using stratified sampling to preserve entity type proportions.

Chapter 3 presented the experimental results and comparative analysis. The key findings are:

1. **Multilingual pre-training is essential.** ruBERT (macro F1 = 0.223) fails catastrophically on the code-switching corpus despite the Russian-dominant character of the text. XLM-RoBERTa (F1 = 0.539) and mE5-base (F1 = 0.608) substantially outperform the monolingual baseline, confirming that massively multilingual encoders are a prerequisite for Kazakh-Russian mixed NLP.

2. **LLM few-shot inference outperforms fine-tuned encoders on overall macro F1.** Qwen2.5:7B achieves the highest macro F1 (0.635) and is the only model to detect `STREET_INTERSECTION` entities (F1 = 0.667). However, its throughput (300 samples in 4+ hours) makes it unsuitable for real-time deployment.

3. **Fine-tuned encoders are superior for high-frequency structured entities.** XLM-RoBERTa and mE5-base achieve `ROUTE_NUM` F1 $\approx$ 0.92, processing thousands of records per minute on GPU. For real-time CTS dispatch triage, XLM-RoBERTa is the recommended production model.

4. `STREET_INTERSECTION` **remains the critical unsolved challenge.** The extreme data scarcity (74 annotated examples) prevents gradient-based models from learning the multi-token conjunctive structure of intersection entities. This is the most important limitation of the current system and the highest-priority target for future data collection.

The contributions of this work are: (1) a custom-annotated bilingual NER corpus for the transport complaint domain, (2) a systematic four-way comparison of NER architectures on Kazakh-Russian code-switching text, (3) entity-level error analysis identifying failure modes specific to this linguistic and domain setting, and (4) practical deployment recommendations for a real-time geotag recognition pipeline.

Future research directions include: targeted data augmentation for rare entities using synthetic data generated from city street graph databases; back-translation between Kazakh and Russian to increase example diversity; hybrid architectures combining a fine-tuned encoder for structured identifiers with an LLM for complex geographic spans; and extension of the pipeline to coordinate resolution through integration with the Astana municipal geocoder API.

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